

Multi-Hop Retrieval-Augmented Generation as Latent Evidence-Chain Inference: A Survey

YUQING LUO, University of Science and Technology of China, China

KAI ZHANG, University of Science and Technology of China, China

Multi-hop retrieval-augmented generation (RAG) must recover an evidence chain whose later units become identifiable only after intermediate states are resolved. A system, however, observes only a noisy retrieval pool, a budgeted context, an evidence ordering, and a generated answer. Existing surveys organize this field primarily by architecture; such taxonomies catalogue systems effectively but do not isolate why they fail. We recast multi-hop RAG as latent evidence-chain inference under missingness, noise, budget, exposure, fusion, and faithfulness constraints. Along a statistical axis, we decompose grounded success into five conditional stage factors: chain observability, selection preservation, exposure probability, fusion reliability, and causal faithfulness. This yields an exact chain-rule factorization and a conditional bottleneck bound, while separating grounded success from shortcut answers and parametric leakage. Conditional evidence utility is treated as a mechanism-level score for improving selection rather than as an additional probability factor. Along a complementary architectural axis, we review retrieval, graph, decomposition, reader, agentic, long-context, multimodal, and mechanistic approaches through 2026. We then reinterpret benchmarks as diagnostic instruments, align empirical claims with conditional subsets and budgets, and propose an auditable reporting standard. A versioned companion artifact separates the broad imported literature library from the reviewed method and benchmark seeds, making the scope and maturity of the released taxonomy explicit.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; **Information extraction**; • **Information systems** → **Question answering**.

Additional Key Words and Phrases: Multi-hop reasoning, retrieval-augmented generation, latent evidence chains, evidence retrieval, evidence fusion, faithfulness, causal attribution, large language models, knowledge graphs, agentic search, process supervision

ACM Reference Format:

Yuqing Luo and Kai Zhang. 2026. Multi-Hop Retrieval-Augmented Generation as Latent Evidence-Chain Inference: A Survey . *ACM Comput. Surv.* 1, 1 (July 2026), 53 pages.

1 INTRODUCTION

Modern information retrieval systems are increasingly asked to answer complex questions whose evidence is distributed across many sources. A multi-hop retrieval-augmented generation (RAG) system is therefore rarely asked to retrieve a single fact; it is asked to recover an unobserved support structure. For a question q , the correct answer depends on a latent evidence chain $C^* = (e_1^*, \dots, e_m^*)$ whose elements must be discovered, ordered, and composed, while the system observes only a noisy retrieval pool L_N , a budgeted context $E_K \subseteq L_N$, an ordering $\pi(E_K)$, and the answer distribution induced by a reader. This mismatch between the latent object required for correctness and the observable objects available to the system is the central statistical difficulty of multi-hop RAG. Figure 1 summarizes the latent target, observable pipeline, and five stage diagnostics.

Consider a question whose answer requires resolving an unstated bridge entity before the answer-bearing evidence can even be located. A single-shot retriever sees only the surface form of the original query and may retrieve highly relevant first-hop documents while missing the bridge document needed for the second hop. Increasing retrieval depth can make the full chain observable but inflates distractor mass; selecting a small context reduces noise but may discard a low-marginal-relevance bridge; reordering the selected evidence can make the chain easier for the

Authors' addresses: Yuqing Luo, University of Science and Technology of China, Hefei, Anhui, China; Kai Zhang, University of Science and Technology of China, Hefei, Anhui, China.

Latent evidence-chain inference view

Observable RAG objects are noisy projections of an unobserved support chain.

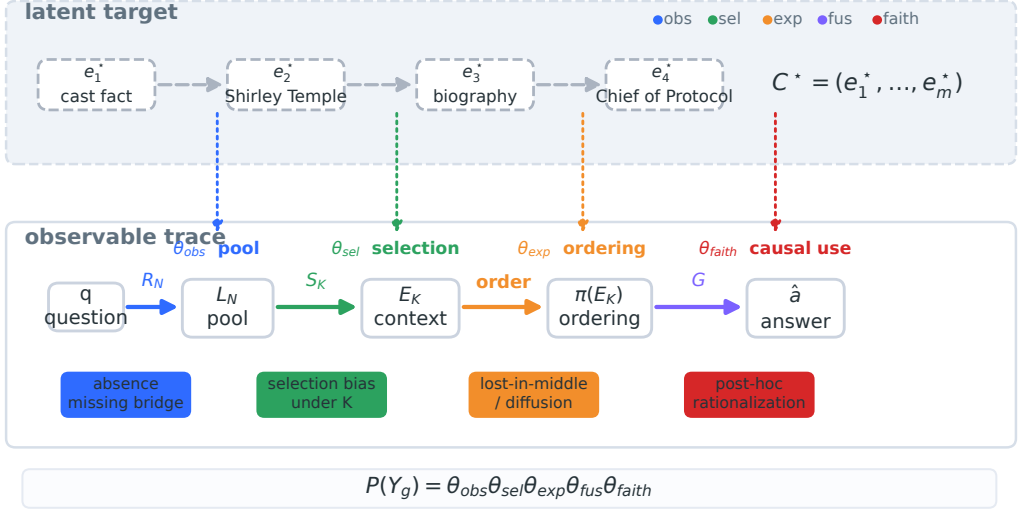


Fig. 1. Latent evidence-chain inference view of multi-hop RAG. The target object is an unobserved support chain C^* , while the system observes only a retrieved pool L_N , a budgeted context E_K , an ordering $\pi(E_K)$, and a generated answer $\hat{a}$. The five estimands diagnose where the chain becomes absent, dropped, poorly exposed, incorrectly fused, or used only post hoc rather than causally.

reader to use but cannot recover evidence that was never selected. Even when the chain is present and visible, a fluent reader may answer from parametric memory or produce a plausible-looking rationale that is not causally responsible for the answer. These are not isolated engineering issues; they are successive statistical bottlenecks in the estimation and use of a latent support chain.

Existing surveys usually organize this field by architectural community: graph-based reading comprehension, iterative retrieval, question decomposition, fusion-in-decoder readers, chain-of-thought prompting, and agentic search [26, 36, 93, 109, 115, 131, 151]. This organization helps locate papers but offers little explanation of why systems fail. Two methods with different surface architectures may target the same statistical bottleneck; conversely, two methods in the same family may fail for different statistical reasons. An iterative retriever, a graph expander, and an agentic search policy all attempt to increase the probability that the latent chain is observable. A reranker, a bridge selector, and a compression model all estimate some version of conditional evidence utility. Verification and counterfactual attribution, although discussed under distinct labels, both target the causal faithfulness of the final answer.

This survey therefore asks a different organizing question: *which quantity of the latent evidence-chain inference problem does a method estimate?* The goal is not to discard architectural taxonomies but to place them under a more diagnostic statistical layer. We use five estimands as primary coordinates: chain observability, selection preservation under budget, exposure probability, fusion reliability, and causal faithfulness. Conditional evidence utility is treated as a mechanism-level selector score rather than as a sixth estimand. These quantities convert common multi-hop failure modes into measurable events and make design tradeoffs explicit. To preserve continuity with the

existing literature, Section 6 provides a parallel architectural review that catalogues methods by their surface design and maps each family to the estimands it primarily affects.

1.1 A Running Example through the Latent-Chain Lens

To make the distinction concrete, we adopt a canonical bridge-type question from the HotpotQA benchmark [212], whose gold supporting facts are annotated and publicly verifiable:

“What government position was held by the woman who portrayed Corliss Archer in the film Kiss and Tell?”

A surface retriever conditioned on the literal query is drawn toward passages about the 1945 film *Kiss and Tell*, but the latent support chain contains at least two dependent units: e_1^* , the bridge-entity unit establishing that Shirley Temple portrayed Corliss Archer in *Kiss and Tell*; and e_2^* , the answer-bearing unit stating that Shirley Temple subsequently served as United States Chief of Protocol. The second unit is not directly queryable from the surface form of the question; it becomes reachable only after the bridge entity has been resolved. We use this canonical two-hop example throughout the survey. When discussing longer dependency chains, we use MuSiQue [167] as a benchmark family rather than introducing an additional untracked running item. In Section 3 the example grounds the formalism $(C^*, L_N, E_K, \pi(E_K), \hat{a})$; in Section 4 it illustrates each of the five estimands; in Sections 5 and 6 it shows how different method families intervene at different points; in Section 7 it anchors the discussion of which benchmarks expose which failure events; and in Section 9 it supports counterfactual faithfulness illustrations.

In the notation used in this survey, the complete support chain C^* is not a document list returned by a system; it is the unobserved object that would make the answer identifiable. The retrieved pool L_N is an imperfect observation of that object, the selected context E_K is a budgeted estimator, and the reader-facing ordering $\pi(E_K)$ is a presentation of the estimator. This separation matters because different failures produce the same final symptom. If the answer is wrong, it may be because the Shirley Temple biography passage was absent from L_N , because it was dropped when E_K was selected under a small context budget, because it was placed far from the *Kiss and Tell* cast passage in $\pi(E_K)$, because the reader failed to compose the two passages, or because the model answered from memorized knowledge without using the chain. Final answer accuracy alone cannot distinguish these cases.

This example also clarifies why the statistical view is not merely a renaming of a pipeline. The latent chain induces a dependency structure, whereas the observed documents are sampled, selected, and ordered under system-specific policies. A retrieval module, a graph module, and an agentic search policy can therefore be compared by the same question: how much do they increase the probability that C^* becomes observable and usable under a fixed budget?

1.2 Position Relative to Existing Surveys

Prior surveys have covered general RAG [26, 36, 58, 86, 231], multi-hop QA [109, 115], complex KBQA [42, 80], knowledge-graph reasoning with LLMs [122, 127], chain-of-thought and step-by-step reasoning [17, 131, 134], hallucination [57, 227], faithfulness and attribution [36, 85], and agentic RAG [93, 151, 193]. A particularly close recent work is the four-axis framework for retrieval-reasoning processes in multi-hop QA [64], which organizes systems by execution plan, index structure, next-step control, and stopping criterion. These surveys are complementary to ours but answer a different question. Their primary axes are typically retrieval mechanism, system architecture, reasoning style, data source, control flow, or application domain. The unit of analysis is procedural: *what does the system do at each step?* We add an estimand-based axis whose unit of analysis is statistical: *which latent quantity does the step attempt to estimate or improve?*

Two procedures with different executions can target the same estimand, and one procedure can target several at once. An agentic loop may simultaneously increase pool observability, estimate conditional evidence utility, and improve faithfulness through critique; conversely, two iterative retrieval systems may both use repeated search but differ sharply in whether they optimize bridge observability or process faithfulness. The two views thus support different reviewer questions: a procedural view asks whether a method has a clear plan, index, trigger, and stopping rule; a latent-chain view asks whether the method improves $P(C^* \subseteq L_N)$, preserves the chain under budget, exposes it early, composes it reliably, or uses it causally. We argue that a mature multi-hop RAG evaluation should report both, and this survey is structured so that the two views appear side by side. The individual ingredients of our framing are not new in isolation: retrieved documents have been treated as latent variables in end-to-end RAG [48, 83]; multi-hop QA has long used evidence and reasoning chains [55, 167, 212]; and recent attribution work connects faithfulness to causal evidence use [98, 192, 205]. Our contribution is to connect these ingredients into a single statistical organization of multi-hop RAG and to use that organization to support a comprehensive literature review whose method descriptions remain familiar to readers used to architectural categories.

Table 1 makes this positioning concrete by tabulating which analytical axes each recent survey adopts. Existing surveys already cover retrieval mechanisms, architectural taxonomies, reasoning styles, and process control comprehensively. Two axes, however, are largely absent from prior work and are the primary contribution of the present survey: a statistical *estimand axis* that assigns each method to the latent quantity it estimates or improves, and a matching *estimand-aligned reporting standard* that ties reported metrics to those quantities. A causal reading of faithfulness (as counterfactual chain use rather than as citation overlap) also appears only sporadically in prior surveys.

1.3 Contributions

This survey makes four contributions.

- **A latent evidence-chain formalism.** We recast multi-hop RAG as inference over an unobserved support chain C^* , with retrieved pools, selected contexts, evidence orderings, and generated answers treated as noisy observations or budgeted estimators of different aspects of that chain. We give both a latent-variable generative formulation and a Markov decision process view that subsumes recent agentic and adaptive systems.
- **An estimand-based decomposition with diagnostic bounds.** We introduce five statistical estimands—chain observability, selection preservation under budget, exposure probability, fusion reliability, and causal faithfulness—and derive an exact chain-rule decomposition of grounded success over their conditional factors. A bottleneck upper bound shows that grounded success cannot exceed the smallest conditional stage-survival factor. Conditional evidence utility remains a mechanism-level scoring quantity for improving selection preservation, not a probability factor in the decomposition. We also state identifiability conditions linking observable diagnostics to latent events.
- **A dual-axis review with auditable artifact boundaries.** We review methods, benchmarks, and analyses along two complementary axes: an estimand axis that identifies what statistical quantity each method targets, and an architectural axis (Section 6) that catalogues methods by graph reasoning, retrieval pipelines, question decomposition, fusion readers, LLM reasoning, agentic search, and hybrid LLM-KG approaches. To avoid inflating coverage, Section 2 distinguishes the manuscript bibliography, the broad imported working library, and the smaller reviewed catalog seed; only the last supports released taxonomy-level judgments. Coverage extends through 2024–2026 advances including process-supervised retrieval, RL-based search

Table 1. Coverage of analytical axes across recent surveys of retrieval-augmented and multi-hop reasoning. A checkmark (✓) indicates that the axis is a primary organizational axis of the survey; a circle (○) indicates partial or secondary treatment; a blank cell indicates that the axis is not addressed. “Estimand axis” and “Reporting standard” isolate the two axes on which the present survey differs from all prior work.

Survey	Scope	Retrieval mech.	Architecture	Reasoning style	Process control	Benchmarks	Faithfulness (causal)	Estimand axis	Reporting standard
Gao et al. [36]	General RAG	✓	✓			✓			
Fan et al. [26]	RAG + LLMs	✓	✓	○		✓			
Huang and Huang [58]	RAG architectures	✓	✓		○	✓			
Zhao et al. [231]	RAG for AIGC	✓	✓	○		○			
Li et al. [86]	Decoupled RAG	✓	✓		○	○			
Mavi et al. [109]	Multi-hop QA	○	✓	○		✓			
Mohammadi et al. [115]	Multi-hop MRC	○	✓	○		✓			
Lan et al. [80]	Complex KBQA	○	✓	✓		✓			
Pan et al. [127]	LLM + KG	○	✓	✓		○			
Chu et al. [17]	Chain-of-thought			✓	○	○			
Plaat et al. [131]	Multi-step reasoning		○	✓	○	○			
Qiao et al. [134]	LLM reasoning			✓		○			
Huang et al. [57]	Hallucination			○		○	○		
Li et al. [85]	Attribution	○				○	○		○
Singh et al. [151]	Agentic RAG	✓	✓	○	✓	○			
Liang et al. [93]	Reasoning RAG	✓	✓	✓	✓	○	○		
Ji et al. [64]	Retrieval–reasoning process	✓	○	○	✓	○			
This survey	Multi-hop RAG (statistical view)	✓	✓	✓	✓	✓	✓	✓	✓

agents, long-context reading, multimodal multi-hop reasoning, mechanistic interpretation, latent reasoning, and retrieval-side safety.

- **Statistical reinterpretation of benchmarks and open problems.** We reinterpret benchmarks as diagnostic instruments that make different latent events observable, propose a reporting standard that aligns metrics with estimands, and recast common open problems as tradeoffs among missingness, noise, budget, exposure, fusion error, and causal grounding.

Scope. We focus on retrieval-augmented multi-hop reasoning over textual, structured, semi-structured, and hybrid knowledge sources. Single-hop QA, purely parametric reasoning without retrieval, general-purpose agents without retrieval-grounded evidence use, and multimodal multi-hop reasoning are discussed when they clarify the latent-chain view. Mechanistic interpretation and latent reasoning are included in dedicated subsections because they speak directly to whether the chain is even an explicit object in modern LLM-based pipelines.

Organization. The survey is organized into three parts. **Part I (Foundations)** contains the introduction (Section 1), survey methodology (Section 2), the latent-chain problem setup (Section 3), and the statistical decomposition (Section 4). **Part II (Literature Review)** contains the estimand-axis method review (Section 5), the architectural-axis review (Section 6), benchmarks (Section 7), and applications (Section 8). **Part III (Analysis and Outlook)** contains the faithfulness analysis (Section 9), open problems and tradeoffs (Section 10), reporting guidelines (Section 11), and conclusion (Section 12). Appendices supply method-level catalogues, mathematical notes, screening protocol, and a writing checklist.

2 SURVEY METHODOLOGY AND LITERATURE COVERAGE

A survey that proposes a new statistical organization must make its own selection procedure explicit. We screened papers along two dimensions: (i) whether the work directly addresses multi-step evidence dependency, and (ii) which latent-chain bottleneck it primarily targets. The first criterion prevents the survey from becoming a generic RAG or generic LLM-reasoning survey; the second prevents the taxonomy from collapsing into a list of architectural names.

2.1 Search and Inclusion Protocol

The seed pool was assembled from ACL Anthology, OpenReview, arXiv, publisher and venue proceedings pages, and official project repositories, with DBLP and Semantic Scholar used as secondary discovery aids. We used keyword families covering multi-hop question answering, complex question answering, retrieval-augmented generation, iterative retrieval, evidence retrieval, graph reasoning, KGQA, chain-of-thought, agentic search, process supervision, hallucination repair, citation faithfulness, counterfactual evaluation, long-context reading, and mechanistic interpretation of multi-hop reasoning. A paper was included when it satisfied at least one of the following conditions: it proposes a multi-hop method; it introduces a benchmark that exposes multi-step evidence dependency; it analyzes faithfulness or causal support in a retrieval-grounded setting; it is a foundational method that later multi-hop RAG systems reuse as a component; or it provides interpretability evidence about how multi-hop reasoning is implemented inside neural models.

We deliberately exclude papers that are only tangentially related. A single-hop retriever is discussed when it forms the upstream pool for a multi-hop system, but not as a full survey object. A general agent paper is discussed only if tool use is used to improve evidence acquisition, reasoning, or verification. A KG embedding method is discussed only when it functions as part of retrieval, graph expansion, or evidence fusion for multi-hop answering. This scope keeps the survey centered on latent evidence-chain inference rather than on every method that can be placed near RAG.

Candidate papers were assigned a primary architectural family (one of: retrieval, graph/KG, decomposition, fusion reader, LLM reasoning, agentic, hybrid, benchmark, or analysis) and a primary estimand (one of: observability, selection, exposure, fusion, faithfulness, or joint). Conditional evidence utility is recorded as a mechanism-level selection signal rather than as a separate estimand. When a paper targeted multiple estimands, the earliest bottleneck deliberately targeted by its central mechanism was marked as primary and material downstream effects were recorded as secondary. Records are promoted from the imported library to the reviewed seed only after their canonical metadata, source URL, scope, primary label, and at least one pipeline intervention have been checked against the paper or an official project page. The survey is not a PRISMA-style systematic review; it is a curated evidence map, and absence from the catalog must not be interpreted as evidence of irrelevance or low quality.

Artifact accounting. To avoid conflating citation breadth, discovery coverage, and taxonomy verification, we report three non-interchangeable quantities:

N_{bib} .

The number of entries rendered in the manuscript bibliography, including surveyed methods, benchmark papers, foundational components, and auxiliary background works.

N_{lib} .

The number of deduplicated records in the imported working library. These records retain broad adjacent coverage and await per-record scope and metadata verification.

N_{seed} .

The number of method records in the released reviewed catalog seed. Benchmark records and

pipeline-intervention mappings are reported separately because they are different artifact types rather than additional methods.

The values are pinned to the artifact revision cited in Appendix B. The three sets overlap but are not a monotone screening funnel: a citation need not appear in the imported working library, and an imported record is not an endorsed taxonomy placement.

Curation flow. New candidates enter a discovery queue, are deduplicated into the working library, and are promoted to the reviewed seed only after the record-level checks above. Discovery results are never merged into the reviewed catalog automatically. The query families, status definitions, and correction policy are maintained in the companion repository.

2.2 Coverage Balance

The final taxonomy is intentionally balanced across historical and LLM-era work. Early systems such as DRQA [8], GRAFT-NET [155], PULLNET [154], MDR [200], BALEEN [74], FiD [60], REALM [48], DPR [73], and ColBERT [75, 139] establish the retrieval, graph, and fusion primitives. Middle-period systems such as IRCot [168], SELF-RAG [2], FLARE [69], ADAPTIVE-RAG [62], and RAPTOR [140] make retrieval control and context organization explicit. Recent work from 2025–2026—including SEARCH-R1 [70], DEEP-RAG [43], RAG-GYM [198], PAR-RAG [221], DUALRAG [16], VERITAS [205], R1-SEARCHER [152], MULTICUBE-RAG [149]—moves the field toward learned search policies, plan-driven retrieval, process supervision, and faithfulness-aware reward design. We therefore treat LLM-era methods as new instantiations of the same evidence-chain problem, not as a clean break from earlier multi-hop QA.

In parallel, we deliberately include three threads that recent surveys underweight. The first is *long-context reading and retrieval* [4, 68, 97, 140, 217], which changes the granularity of evidence units. The second is *multimodal and table-text multi-hop reasoning* [7, 10, 13, 159, 162, 183], which broadens the substrate of $\mathcal{K}$ beyond passages. The third is *mechanistic interpretation and latent reasoning* [6, 37, 40, 50, 176, 208], which speaks to whether the chain is an explicit object inside the model or an implicit trajectory through its hidden states.

2.3 Positioning Against Recent Process-Centric Surveys

Table 1 summarizes axis coverage across recent surveys. Two entries in that table warrant a closer comparison because they are the nearest neighbours to the present work.

The four-axis framework of Ji et al. [64] organizes multi-hop retrieval–reasoning systems by execution plan, index structure, next-step control, and stopping criterion. Its unit of analysis is procedural: *what the system does at each step*. Ours is statistical: *which latent quantity the step attempts to estimate or improve*. The two axes are orthogonal rather than competing. A process feature such as “trigger retrieval when generation uncertainty exceeds a threshold” can serve different estimands depending on what the triggered retrieval optimizes: deeper bridge coverage (θ_{obs}), selection preservation (θ_{sel}), or faithfulness verification (θ_{faith}). Conversely, the same estimand can be improved by different procedural choices. We therefore adopt the estimand view as the primary axis (Sections 4 and 5) while retaining architectural and procedural positions wherever they help readers trained in prior taxonomies (Section 6).

The reasoning-oriented RAG survey of Liang et al. [93] is the second closest neighbour: it covers agentic control and reasoning styles densely and touches on attribution. It still stops short of a decomposition that attributes end-to-end success to identifiable latent events, and does not propose an estimand-aligned reporting standard. We treat this gap, rather than any incompatibility, as the main opening addressed by the present survey.

Table 2. Literature coverage organized by the primary object optimized. The families overlap; the table describes scope rather than reporting mutually exclusive screening counts.

Family	Core period	Representative role in the survey
Decomposition	2019–2026	Converts a complex query into lower-variance subqueries or executable steps.
CoT / search reasoning	2022–2026	Supplies reasoning topologies and search policies for chain construction.
Verification & faithfulness	2023–2026	Measures or improves step-level support and causal faithfulness.
Retrieval-based RAG	2017–2026	Controls chain observability and evidence utility under budget.
Agentic / RL search	2022–2026	Learns when, what, and how to retrieve in an interactive environment.
KG/ graph reasoning	2018–2026	Represents chains as paths or subgraphs and constrains fusion.
LLM +KG/ GraphRAG	2023–2026	Uses graph structure as a skeleton for LLM reasoning and generation.
Long-context / hierarchical	2023–2026	Reduces the number of explicit hops by changing retrieval granularity.
Multimodal / tabular	2020–2026	Broadens evidence substrates to images, tables, and structured records.
Mechanism / latent reasoning	2023–2026	Reveals or implies chain processes inside model internals.
Benchmarks and analyses	2018–2026	Defines observable proxies for latent chain events.

3 PROBLEM SETUP: FROM RETRIEVED DOCUMENTS TO LATENT EVIDENCE CHAINS

3.1 What Makes a Question Multi-Hop?

A task is *multi-hop* when its solution requires connecting multiple evidence units whose discovery or use depends on intermediate states. Three properties characterize this dependency. First, *semantic non-locality*: at least one required evidence unit is not directly retrievable from the original question because its surface terms are absent. Second, *ordering sensitivity*: evidence must be combined through an ordered computation, such as bridge resolution, filtering, projection, comparison, or aggregation. Third, *joint sufficiency*: no strict subset of the support chain is enough to answer the question.

These properties distinguish multi-hop RAG from adjacent tasks. A long-document task may require high recall but remain single-hop if all relevant sentences are directly query-relevant. A multi-step generation task is not multi-hop unless different steps require different external evidence with cross-step dependency. A hard reasoning task with one piece of evidence is single-hop with difficult inference, not multi-hop retrieval. Returning to the running example, the question is multi-hop because the biography passage stating that Shirley Temple served as U.S. Chief of Protocol is not retrievable from the surface terms of the question about the film *Kiss and Tell*: that biography becomes queryable only after the bridge entity Shirley Temple has been resolved.

3.2 Latent Chain and Observable Variables

Let q be a question and $\mathcal{K} = \{e_1, e_2, \dots\}$ be a knowledge store. An evidence unit may be a passage, sentence, span, table cell, KG triple, or path. We denote the unobserved support chain required to answer q by

$$C^*(q) = (e_1^*, \dots, e_m^*), \quad (1)$$

where the order encodes dependency among evidence units. The correct answer is a^* .

A multi-hop RAG system does not observe C^* directly. It instead produces a sequence of observable objects:

$$L_N = \mathcal{R}_N(q), \quad (2)$$

$$E_K = S_K(q, L_N), \quad E_K \subseteq L_N, \quad K \ll N, \quad (3)$$

$$\pi(E_K) = \text{an ordering or presentation of } E_K, \quad (4)$$

$$\hat{a} = \mathcal{G}(q, \pi(E_K)). \quad (5)$$

Here L_N is the retrieved pool, E_K is the selected budgeted context, $\pi(E_K)$ is the reader-facing ordering, and $\hat{a}$ is the answer. This sequence defines the observable side of the latent-chain inference problem. For the running example, C^* has two ordered units (film-cast passage identifying Shirley Temple as the actress who portrayed Corliss Archer $\rightarrow$ biography passage stating her later role as U.S. Chief of Protocol); L_N might contain the film-plot summary, several biography passages of other cast members, distractors about unrelated Shirley Temple films, and possibly the answer-bearing biography passage; E_K is the top- K subset that the selector keeps; $\pi(E_K)$ determines whether the film-cast passage is placed before or after the biography passage.

3.3 Failure Modes as Latent-Event Failures

The usual failure modes of multi-hop RAG can be written as failures of latent events:

Evidence absence.

$C^* \not\subseteq L_N$. A required unit is missing from the retrieved pool, often at the bridge hop.

Budgeted selection failure.

$C^* \subseteq L_N$ but $C^* \not\subseteq E_K$. The chain is observable in the pool but is not preserved under a small context budget.

Exposure failure.

$C^* \subseteq E_K$ but the chain is placed too late or too diffusely in $\pi(E_K)$ for the reader to use reliably. This includes lost-in-the-middle [97] and ordering-gap effects.

Fusion or reasoning failure.

The chain is selected and visible, but the reader fails to compose the evidence units into the answer.

Faithfulness failure.

The answer is correct or plausible but is not causally supported by the retrieved chain. This includes incorrect citations and parametric leakage.

This event view is stricter than a natural-language taxonomy: it asks where, in the observable sequence $L_N \rightarrow E_K \rightarrow \pi(E_K) \rightarrow \hat{a}$, the latent support chain becomes unavailable, unusable, or non-causal. In the running example, “evidence absence” might be a missing Shirley Temple biography passage; “budgeted selection failure” might be the biography passage being ranked below high-surface but unrelated film-plot documents; “exposure failure” might be the answer-bearing biography placed in position eight of an eight-passage prompt; “fusion failure” might be that the reader connects Shirley Temple to the wrong later career role; and “faithfulness failure” might be that the reader answers “U.S. Chief of Protocol” correctly because that fact is memorized from pretraining, while the cited chain does not actually entail the answer.

3.4 Adjacent Problems

The latent-chain view also clarifies what is *not* multi-hop RAG. Several adjacent research problems share surface machinery with multi-hop RAG—retrieval over a corpus, multi-step generation, compositional reasoning, or grounded question answering—but differ in whether the target object

Table 3. Multi-hop RAG and adjacent research problems under the latent-chain view.

Adjacent problem	Shared aspect	Distinguishing aspect of multi-hop RAG
Single-hop open-domain QA	Retrieval over an external corpus	The latent object is a chain, not one directly retrievable fact.
Chain-of-thought reasoning	Multi-step intermediate text	Intermediate steps must be grounded in retrievable evidence, not only model-internal computation.
Compositional generalization	Combining components	Composition occurs over distributed evidence units, not only over latent functions.
KG path reasoning	Sequential relation traversal	A KG path is one possible chain substrate; text, tables, and hybrid stores also instantiate $\mathcal{K}$.
General hallucination detection	Detecting unsupported claims	Faithfulness must hold at every chain step, not only at the final claim.
Agentic LLM systems	Tool use and planning	Actions are evaluated by whether they improve latent-chain inference.
Long-context reading	Conditioning on long input	Granularity of evidence units shifts, but the chain may still be non-local and bridge-dependent.
Multi-document summarization	Combining many documents	Output is open-ended summary rather than a chain-identifiable answer.

is a dependency-linked chain of evidence units. We summarize these distinctions in Table 3. Two axes matter: whether the task’s correctness object is a chain rather than a single fact or a free-form summary, and whether intermediate steps must be grounded in retrievable evidence rather than only in model-internal computation. Adjacent problems typically share one of these axes but not both; methods designed for them can supply useful components (retrievers, verifiers, planners) but do not, on their own, address the joint latent-chain estimation problem that this survey studies.

3.5 Reasoning Types as Chain Structures

The latent-chain view gives a compact way to describe common multi-hop reasoning types. *Bridge reasoning* corresponds to a chain in which an early evidence unit resolves an unmentioned entity needed to retrieve a later unit. *Comparison reasoning* corresponds to two or more parallel subchains whose terminal attributes must be compared. *Compositional reasoning* corresponds to a sequence of operators such as filter, project, aggregate, and compare applied over distributed evidence [161, 167]. *Verification-oriented reasoning* corresponds to a chain whose terminal object is not an answer string but a support or refutation label [67, 102, 165]. *Counterfactual probing* replaces one node or relation in C^* and tests whether the induced answer distribution changes [192].

These reasoning types differ in the shape of C^* . A bridge question is often close to a path; a comparison question is closer to a small tree; a fact-verification problem may require a set of jointly supporting facts; and a table-text question may alternate between cells, linked passages, and entity mentions [10, 13]. The statistical bottlenecks remain the same. The system must make the relevant structure observable, select it under budget, expose it to the reader, compose it correctly, and use it causally. This is why the same estimand-based vocabulary can cover graph paths, text passages, tables, and agent trajectories.

3.6 A More Explicit Generative View

The preceding notation can be written as a latent-variable model. Let $\mathfrak{C}(q)$ denote the set of admissible evidence chains for q . A dataset or world distribution first samples a valid chain $C \in \mathfrak{C}(q)$,

then an answer is generated from the chain:

$$C \sim p(C \mid q, \mathcal{K}), \quad (6)$$

$$a^* \sim p(a \mid q, C). \quad (7)$$

A retrieval-augmented system observes only a corrupted view of this chain through retrieval and context construction:

$$L_N \sim p_R(L \mid q, \mathcal{K}), \quad (8)$$

$$E_K \sim p_S(E \mid q, L_N, K), \quad (9)$$

$$\pi \sim p_O(\pi \mid q, E_K), \quad (10)$$

$$\hat{a} \sim p_G(a \mid q, \pi(E_K)). \quad (11)$$

Thus the ideal Bayesian objective is

$$\hat{a}_{\text{Bayes}} = \arg \max_a \sum_{C \in \mathfrak{C}(q)} p(a \mid q, C) p(C \mid q, \mathcal{P}_{\text{sys}}), \quad (12)$$

where $\mathcal{P}_{\text{sys}} = (L_N, E_K, \pi(E_K))$ are the observations made available by the system. Real systems approximate this posterior through retrieval scores, rerankers, graph traversals, prompts, tool calls, and verifiers. The statistical question is how each approximation changes the posterior mass assigned to complete and causally sufficient chains.

Definition 1 (Minimal support chain). For a question q and answer a^* , a chain $C = (e_1, \dots, e_m)$ is *sufficient* if $p(a^* \mid q, C)$ exceeds a task-specific correctness threshold. It is *minimal* if no proper subsequence of C is sufficient. The set of all minimal chains is denoted by $\mathfrak{C}^*(q)$.

This definition is deliberately set-valued. Many questions admit multiple valid chains. A Wikipedia biography, a table row, and a KG path may all support the same answer. Treating C^* as a single sequence is therefore a notational convenience; the more faithful object is a set or distribution over minimal chains. When the survey writes $C^* \subseteq L_N$, the statement should be read as “there exists at least one valid minimal chain fully contained in the pool.”

Definition 2 (Multi-hop dependency graph). Given a minimal chain $C = (e_1, \dots, e_m)$, define a directed graph $D_C = (V, A)$ whose vertices are evidence units and whose edge $e_i \rightarrow e_j$ indicates that e_j becomes identifiable, retrievable, or interpretable only after information in e_i is known. The hop count is the length of the longest directed path in D_C .

This graph definition clarifies why hop count is not identical to the number of retrieved passages. A comparison question may have two parallel single-hop branches but still be multi-hop because the final comparison operator depends on both branches. A bridge question may have a path-like structure. A table-text question may alternate between table cells and linked passages. The chain formalism abstracts over these substrates.

Assumption 1 (Budgeted observability). For a fixed final context budget K , a multi-hop system can answer only if at least one minimal chain has cardinality at most K after the chosen evidence-unit granularity is fixed.

Assumption 1 is not a statement about the world; it is a statement about an evaluation protocol. If the minimal chain requires six passages but the reader receives only five, then no selector can preserve a passage-level chain. The only possible remedies are to change the granularity, compress evidence, use a larger budget, or perform iterative generation with additional retrieval calls.

3.7 State-Space and MDP Formulations

Many recent agentic and adaptive RAG methods can be written as a controlled stochastic process [2, 43, 62, 70, 198]. Let $s_t = (q, h_t, L_t, M_t)$ be the state at step t , where h_t is the reasoning history, L_t is the accumulated retrieval pool, and M_t is a memory or scratchpad. An action u_t may be a query rewrite, retrieval call, graph expansion, compression step, verification call, or final-answer action. The transition is

$$s_{t+1} \sim P(s_{t+1} \mid s_t, u_t, \mathcal{K}), \quad (13)$$

and the terminal reward can combine correctness, faithfulness, and cost:

$$R_T = \mathbb{I}\{\hat{a} = a^*\} - \lambda_c \sum_{t=0}^{T-1} \text{cost}(u_t) - \lambda_f \mathbb{I}\{\hat{a} \leftarrow C^*\}. \quad (14)$$

This casts agentic RAG as policy learning:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi, P} [R_T]. \quad (15)$$

DEEPRAG [43], SEARCH-R1 [70], R1-SEARCHER [152], RESEARCH [9], and RAG-GYM [198] differ mainly in what state variables are observable, how actions are parameterized, whether rewards are outcome-only or process-supervised, and how retrieved tokens are handled during optimization.

The MDP view is useful but incomplete. The true support chain is still latent; the policy receives sparse and biased observations. Therefore, agentic search does not remove the latent-chain problem. It merely moves the approximation from a fixed retriever to a learned control policy over retrieval and reasoning actions.

3.8 Where Does the Chain Live? Latent Reasoning and Mechanistic Interpretation

A separate but related thread asks whether the chain is even an explicit object inside modern LLM-based systems. *Latent reasoning* methods compress some or all reasoning steps into hidden states rather than tokens. QUIET-STAR [40] trains models to silently generate internal rationales before each token. COCONUT [50] performs continuous chain-of-thought in a latent space, decoupling reasoning from natural language. Pause tokens and “think before speak” variants insert non-emitted computation [39]. From the latent-chain view, these methods do not eliminate C^* ; they make it unobservable through tokens, which complicates both diagnostic evaluation and faithfulness attribution.

Mechanistic interpretation of multi-hop reasoning takes the opposite stance: it locates traces of C^* inside model internals. Probing work has shown that LLMs sometimes recall bridge entities in intermediate layers [6, 37, 208], that the recall of a second-hop fact can be circuit-localized [176], and that some questions are answered through “shortcut” rather than genuine bridge recall [208]. These findings matter for this survey for two reasons. First, they suggest that even when no chain is written into the prompt, an implicit chain can sometimes be recovered from internal activations. Second, they motivate diagnostics that combine retrieved-evidence sensitivity with internal-state sensitivity: a model whose answer changes when an internal bridge representation is patched is more plausibly using the chain causally than one whose answer is invariant under such interventions.

We return to this dimension in Section 9 when discussing faithfulness diagnostics, and in Section 10 when discussing open problems for latent reasoning under retrieval augmentation.

4 A STATISTICAL DECOMPOSITION OF MULTI-HOP RAG

This section introduces the five statistical quantities used to organize the rest of the survey. Each quantity corresponds to one stage at which a latent support chain can be lost or misused.

4.1 Chain Observability

The first question is whether the complete support chain appears in the retrieved pool:

$$\theta_{\text{obs}}(q, N) = P(C^* \subseteq L_N \mid q). \quad (16)$$

This quantity captures retrieval-side missingness. If θ_{obs} is low, no selector, reader, or verifier can recover a reliable chain from the pool. When a chain has m hops and each required unit has marginal retrieval probability p_i , then under an independence approximation,

$$P(C^* \subseteq L_N) \approx \prod_{i=1}^m p_i. \quad (17)$$

This explains a basic scaling law of multi-hop difficulty: even modest per-hop miss rates compound rapidly as the chain length grows. Empirically, on MuSiQUE [167] 4-hop subsets, many single-shot retrievers achieve high recall on the first hop but nearly zero pool-level chain recall, exactly as the product form predicts.

For the running example, θ_{obs} depends on whether both the *Kiss and Tell* cast passage and the Shirley Temple biography passage appear in L_N . A single-shot dense retriever conditioned only on the surface form of the question is drawn toward film-plot passages and is unlikely to surface the biography of the bridge entity; the conditional probability of retrieving the answer-bearing biography rises only after Shirley Temple has been resolved as the bridge.

4.2 Conditional Evidence Utility and Selection Preservation

Once the chain is observable in L_N , two distinct quantities govern what happens next: (i) a *mechanism* quantity that a selector uses to rank candidate units against a partial chain, and (ii) an *event* quantity that records whether the full chain survives budgeted selection. Conflating the two is a common source of confusion in the multi-hop literature; we therefore separate them.

Conditional evidence utility (mechanism quantity). Standard retrievers estimate marginal relevance $s_{\text{rel}}(q, e)$, but multi-hop selection requires

$$\theta_{\text{util}}(e \mid q, C) = P(a^* \mid q, C \cup \{e\}) - P(a^* \mid q, C), \quad (18)$$

which takes values in $[-1, 1]$ and measures how much adding e to a partial context C changes the answer probability. This quantity is a signed *marginal contribution*; it is not itself a probability. Bridge documents often look weak under query-only relevance but have strongly positive conditional utility once a partial chain is known, and distractors can have negative utility. Reranking, bridge selection, evidence compression, and core-conditioned selection all approximate θ_{util} or aggregates of it [74, 201, 220, 221].

Selection preservation (event quantity). Whether the selector, guided by any such mechanism, actually keeps the support chain under a small budget K is a separate probabilistic event. Given that the chain is present in the pool ($C^* \subseteq L_N$), let

$$\theta_{\text{sel}} = P(C^* \subseteq E_K \mid C^* \subseteq L_N) \in [0, 1]. \quad (19)$$

This is the quantity that appears as a factor in the success decomposition of Section 4.6 and in Proposition 1. It is a genuine conditional probability, distinct from the mechanism quantity above. In practice, θ_{sel} is what a Bridge-Recall or Chain-Preservation metric measures on a held-out set, whereas θ_{util} is what a per-candidate rerank score approximates.

4.3 Exposure Probability

Selection is not enough. A reader with limited attention or positional bias may fail to use a complete chain if the relevant units are late, scattered, or separated by distractors. We define exposure probability as

$$\theta_{\text{exp}}(q, K, r) = P(\text{rank}_{\pi}(C^{\star}) \leq r \mid C^{\star} \subseteq E_K), \quad (20)$$

where $\text{rank}_{\pi}(C^{\star})$ denotes the latest or most difficult-to-access position of the required chain under ordering π , and r is a reader-dependent “effective access depth” beyond which positional bias starts to dominate [97]. Ordering, reader-aware reranking, context packing, and compression methods target this quantity.

4.4 Fusion Reliability

When the chain is present and exposed, the reader must compose it. We define fusion reliability as

$$\theta_{\text{fus}}(q) = P(\hat{a} = a^{\star} \mid C^{\star} \subseteq E_K, \text{rank}_{\pi}(C^{\star}) \leq r). \quad (21)$$

This quantity separates retrieval success from reasoning success. It explains why high evidence recall does not imply high answer accuracy: the reader may fail to compare, aggregate, bridge, or verify the selected evidence.

4.5 Causal Faithfulness

Finally, correctness does not imply faithful evidence use. We treat causal faithfulness as the latent event that the generated answer actually depends on a valid support chain, rather than merely co-occurring with plausible evidence or being reconstructed from parametric memory. Conceptually, let $\hat{a} \leftarrow C^{\star}$ denote this causal-grounding event and define

$$\theta_{\text{faith}}(q) = P(\hat{a} \leftarrow C^{\star} \mid \hat{a} = a^{\star}). \quad (22)$$

This event is not directly identifiable from answer and citation outputs alone. We therefore use interventional diagnostics as observable proxies rather than definitions of faithfulness itself.

One such proxy is chain-deletion sensitivity. Let $\mathcal{G}(q, \pi(E_K))$ denote the reader’s induced answer distribution. For any candidate support chain $C \subseteq E_K$, define

$$D_{\text{del}}(C) = \Pr[\mathcal{G}(q, \pi(E_K)) = a^{\star}] - \Pr[\mathcal{G}(q, \pi(E_K) \setminus C) = a^{\star}]. \quad (23)$$

For a fixed threshold $\tau > 0$, the indicator $\mathbf{1}\{D_{\text{del}}(C) \geq \tau\}$ is an operational proxy for whether C is load-bearing for the observed answer. It is neither necessary nor sufficient for causal faithfulness: deletion can create out-of-distribution inputs, redundant or alternative valid chains can make a faithful answer deletion-insensitive, and a shortcut can remain sensitive to superficial context changes. Accordingly, a credible faithfulness audit should combine deletion with counterfactual bridge substitution, citation entailment, and closed-book or conflict controls.

The conceptual faithfulness gap is

$$\Delta_{\text{faith}} = P(\hat{a} = a^{\star}, \hat{a} \nleftarrow C^{\star}), \quad (24)$$

but no single deletion statistic is assumed to upper-bound this mass. Verification, citation checking, counterfactual evaluation, and process-supervised reward models provide complementary evidence about this gap [32, 38, 95, 158, 192, 205, 207, 232].

4.6 Success-Factor Decomposition

We define five cumulative latent events along the observable pipeline $L_N \rightarrow E_K \rightarrow \pi(E_K) \rightarrow \hat{a}$:

$$\mathcal{P} = \{C^\star \subseteq L_N\}, \quad (25)$$

$$\mathcal{S} = \mathcal{P} \cap \{C^\star \subseteq E_K\}, \quad (26)$$

$$\mathcal{O} = \mathcal{S} \cap \{\text{rank}_\pi(C^\star) \leq r\}, \quad (27)$$

$$\mathcal{U} = \mathcal{O} \cap \{\mathcal{G} \text{ correctly composes } C^\star\}, \quad (28)$$

$$\mathcal{T} = \mathcal{U} \cap \{\hat{a} = a^\star \text{ and is causally supported by } C^\star\}. \quad (29)$$

The intersections make the nesting explicit: $\mathcal{T} \subseteq \mathcal{U} \subseteq \mathcal{O} \subseteq \mathcal{S} \subseteq \mathcal{P}$. We define the *grounded-success* event as their intersection,

$$Y_g = \mathcal{P} \cap \mathcal{S} \cap \mathcal{O} \cap \mathcal{U} \cap \mathcal{T}, \quad (30)$$

and write ordinary answer success as $Y = \{\hat{a} = a^\star\}$, so that

$$P(Y) = P(Y_g) + P(Y \cap \neg Y_g), \quad (31)$$

where the second term captures shortcuts, parametric leakage, and lucky guesses in which the answer is correct without a full grounded chain. Under the nested structure above, the chain rule yields the *exact* factorization

$$P(Y_g) = P(\mathcal{P}) P(\mathcal{S} \mid \mathcal{P}) P(\mathcal{O} \mid \mathcal{S}) P(\mathcal{U} \mid \mathcal{O}) P(\mathcal{T} \mid \mathcal{U}). \quad (32)$$

We denote the five conditional factors by $\theta_{\text{obs}} = P(\mathcal{P})$, $\theta_{\text{sel}} = P(\mathcal{S} \mid \mathcal{P})$, $\theta_{\text{exp}} = P(\mathcal{O} \mid \mathcal{S})$, $\theta_{\text{fus}} = P(\mathcal{U} \mid \mathcal{O})$, and $\theta_{\text{faith}} = P(\mathcal{T} \mid \mathcal{U})$; these are the estimands whose empirical estimators we survey in Section 5. Figure 2 visualizes the conditional factorization, its active bottleneck, and ordinary answer accuracy.

Remark 1 (On the meaning of equation 32). Equation 32 is an exact chain-rule decomposition of the grounded-success event Y_g and does not assume conditional independence. It is a *diagnostic* factorization in the sense that each conditional factor is empirically estimable on the subset of instances that reach the preceding event: θ_{sel} on instances with $\mathcal{P}$, θ_{exp} on instances with $\mathcal{S}$, and so on. Because real systems admit shortcut and parametric-leakage events, $P(Y)$ can exceed $P(Y_g)$; the gap $P(Y) - P(Y_g)$ is itself a diagnostic quantity and is what faithfulness-oriented evaluation (Section 9) seeks to expose.

Proposition 1 (Conditional bottleneck bound). *Let θ_i denote the five conditional stage factors in equation 32. Then*

$$P(Y_g) = \prod_i \theta_i \leq \min_i \theta_i. \quad (33)$$

If the minimum factor is positive, equality requires every other factor to equal one; if any factor is zero, both sides are zero. Thus grounded success cannot exceed the weakest conditional stage-survival rate.

PROOF. Every θ_i lies in $[0, 1]$. For any index k , $\prod_i \theta_i = \theta_k \prod_{j \neq k} \theta_j \leq \theta_k$; taking the minimum over k proves the bound. The equality conditions follow directly from the product of the remaining factors. $\square$

A coordinate view of local sensitivity. The product of the five conditional factors is exactly $P(Y_g)$ by equation (32). What is heuristic is treating those factors as independently perturbable coordinates. Define

$$\Phi(\theta) := \theta_{\text{obs}} \theta_{\text{sel}} \theta_{\text{exp}} \theta_{\text{fus}} \theta_{\text{faith}} = P(Y_g). \quad (34)$$

Bottleneck coupling among estimands

A single weak factor caps end-to-end success even when other modules are strong.

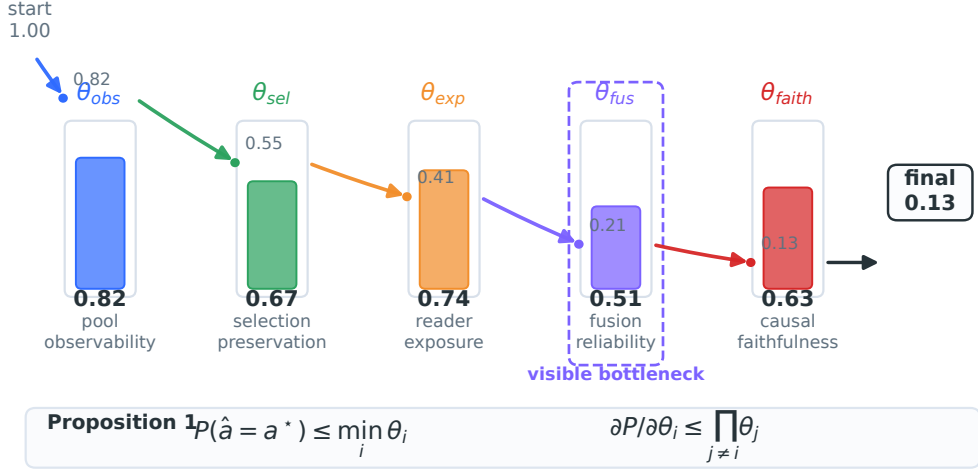


Fig. 2. Bottleneck visualization of the five estimands. Grounded end-to-end success $P(Y_g)$ is upper-bounded by the smallest conditional stage factor across observability, selection preservation, exposure, fusion, and faithfulness. Answer accuracy $P(Y)$ additionally receives a non-grounded contribution from shortcut and parametric-leakage events (equation 31).

The coordinate derivative

$$\frac{\partial \Phi}{\partial \theta_i} = \prod_{j \neq i} \theta_j \quad (35)$$

describes the local change in the product if one factor could be varied while the others were held fixed. The product identity is exact for the conditional factors; only the interpretation of an isolated partial derivative as an achievable system intervention is heuristic, because in real systems changing a retriever, selector, or reader generally changes several conditional distributions at once. We therefore use equation (35) only to summarize coordinate sensitivity, not to bound the true marginal gain of a system update.

Proposition 1 summarizes a direct implication of the conditional factorization: a strong retriever does not improve grounded accuracy when the reader cannot fuse evidence; a strong reader does not help when the chain is absent from the pool; a faithfulness verifier may reject unfaithful answers but cannot recover missing evidence. It also implies that the most credible empirical papers are those that show *where in the product* the gain occurs, and that headline gains in $P(Y)$ without a matching gain in the corresponding component event are candidates for shortcut-driven improvement rather than grounded improvement.

4.7 Risk under Missingness, Noise, and Budget

The estimand view also yields a compact risk perspective. Let ρ_N and ρ_K denote, respectively, the noise ratios of the retrieved pool and the selected context,

$$\rho_N = \frac{|L_N \setminus C^*|}{|L_N|}, \quad \rho_K = \frac{|E_K \setminus C^*|}{|E_K|}, \quad (36)$$

Table 4. Estimand-based taxonomy of multi-hop RAG. Methods are grouped by the statistical quantity they primarily estimate or improve.

Estimand	Mathematical form	Diagnosed bottleneck	Representative mechanisms
Chain observability	$P(C^* \subseteq L_N)$	Evidence missingness	Iterative retrieval, graph expansion, agentic search
Selection preservation	$P(C^* \subseteq E_K \mid C^* \subseteq L_N)$	Budgeted selection bias	Reranking, bridge selection, compression, core-conditioned selection
Exposure probability	$P(\text{rank}_\pi(C^*) \leq r)$	Ordering and lost-in-the-middle	Reader-aware ordering, passage packing, context compression
Fusion reliability	$P(\hat{a} = a^* \mid C^* \subseteq E_K)$	Cross-evidence reasoning failure	FiD, graph fusion, decomposition, programmatic reasoning
Causal faithfulness	$P(\hat{a} \leftarrow C^* \mid \hat{a} = a^*)$	Correct but unsupported answers	Verification, citation checking, counterfactual attribution, PRM

so that ρ_N measures pool-level distractor mass and ρ_K measures how much of the budgeted context is spent on non-chain units. A stylized system risk can be written as

$$\mathcal{R}(N, K) = \alpha P(C^* \not\subseteq L_N) + \beta \mathbb{E}[\rho_K] + \gamma \text{Cost}(N, K) + \delta \Delta_{\text{faith}}, \quad (37)$$

where the expectation in the second term is over questions and retrieval outcomes. This expression makes explicit why multi-hop RAG is not solved by “retrieve more” or “use a larger reader.” Deeper retrieval reduces missingness but increases noise and cost; smaller contexts reduce noise but increase selection bias; stronger readers may improve fusion while leaving faithfulness unresolved.

4.8 Identifiability and Surrogate Supervision

The five estimands are conceptually clean but not always directly observable. In most datasets, C^* is partially annotated, non-unique, or absent. Supporting-fact labels are only a surrogate for chain membership; they may omit alternative valid chains, include facts that are sufficient but not necessary, or differ from the path actually used by the model. This creates an identifiability problem: answer accuracy can be observed, but the latent event that failed is often not identified without additional instrumentation.

Proposition 2 (Conditional identifiability). *Suppose $\mathfrak{C}^*(q)$ is the full set of valid support chains for question q , and that for each example we observe a subset $\Sigma(q) \subseteq \mathfrak{C}^*(q)$ of annotated valid chains and the full observable sequence $(L_N, E_K, \pi(E_K), \hat{a})$. Define the annotated versions of the observability, selection, and exposure events by*

$$\mathcal{P}^\Sigma = \{\exists C \in \Sigma(q) : C \subseteq L_N\}, \quad (38)$$

$$\mathcal{S}^\Sigma = \{\exists C \in \Sigma(q) : C \subseteq E_K\}, \quad (39)$$

$$\mathcal{O}^\Sigma = \{\exists C \in \Sigma(q) : C \subseteq E_K \text{ and } \text{rank}_\pi(C) \leq r\}. \quad (40)$$

Then $\mathcal{P}^\Sigma$, $\mathcal{S}^\Sigma$, and $\mathcal{O}^\Sigma$ are directly identifiable from $(\Sigma(q), L_N, E_K, \pi(E_K))$ and satisfy $\mathcal{P}^\Sigma \subseteq \mathcal{P}$, $\mathcal{S}^\Sigma \subseteq \mathcal{S}$, $\mathcal{O}^\Sigma \subseteq \mathcal{O}$; an observed positive on the annotated event implies the underlying latent event, while an observed negative on the annotated event is informative for the latent event only up to the chain-set gap $\mathfrak{C}^*(q) \setminus \Sigma(q)$. The events $\mathcal{U}$ and $\mathcal{T}$ are not identifiable from $(\Sigma(q), L_N, E_K, \pi(E_K), \hat{a})$ alone; they require interventional probes such as evidence deletion, bridge counterfactual, or oracle-evidence comparison (Section 9).

Table 5. Estimand-level diagnostics for multi-hop RAG. Each metric is best interpreted conditionally on the preceding event.

Stage	Event	Possible diagnostic	Common confounder
Pool construction	$\mathcal{P} = \{C^* \subseteq L_N\}$	pool support recall, path recall, bridge recall	shortcut labels or incomplete gold chains
Budgeted selection	$\mathcal{S} = \{C^* \subseteq E_K\}$	selected support recall, chain preservation rate	relevance-biased gold annotations
Presentation	$\mathcal{O} = \{\text{rank}_\pi(C^*) \leq r\}$	early support exposure, support adjacency, lost-in-middle sensitivity	simultaneous membership changes
Fusion	$\mathcal{U}$	oracle-evidence EM/F1, step accuracy, path validity	parametric answer leakage
Faithfulness	$\mathcal{T}$	citation precision, deletion tests, counterfactual sensitivity	post-hoc rationalization

We distinguish three supervision regimes. In *fully annotated* regimes, path or step labels [18, 55, 167, 212] allow approximate measurement of observability and selection. In *weakly annotated* regimes, supporting facts or citations provide noisy labels for membership and faithfulness. In *unannotated* regimes, the system must rely on proxies such as retrieval overlap, counterfactual perturbation, leave-one-out deletion [98], or verifier scores [205].

These regimes should not be mixed without qualification. A method that improves answer F1 under unannotated evaluation may have improved observability, fusion, or parametric guessing; without estimand-level diagnostics the mechanism is underdetermined. This perspective suggests a reporting standard for future work (Section 11): for any multi-hop system, authors should report at least one metric for each of three stages: pool-level chain coverage, context-level chain preservation, and answer-level faithfulness.

4.9 A Diagnostic Protocol

The statistical decomposition induces a simple diagnostic protocol. First, evaluate the retrieved pool L_N against available support labels to estimate chain observability. If the chain is absent, downstream reader errors should not be attributed to reasoning. Second, condition on chain-present examples and measure whether the selected budgeted context E_K preserves the chain. This isolates selection bias under budget. Third, hold membership fixed and vary the ordering $\pi(E_K)$ to measure exposure effects. Fourth, evaluate the reader under oracle or near-oracle evidence to estimate fusion reliability. Fifth, perturb or remove cited evidence to test causal faithfulness.

The important point is that these diagnostics are conditional. A faithfulness test on examples where the evidence is absent conflates retrieval and attribution. A fusion test on examples where the chain is not selected conflates selection and reasoning. A ranking test that changes both membership and order cannot isolate exposure. Conditional evaluation is therefore not a minor experimental detail; it is required by the latent-chain factorization itself.

4.10 Monotonicity, Interaction, and Submodularity

A common implicit assumption in retrieval pipelines is that evidence utility is approximately additive: if each document is relevant on its own, the set should be useful. Multi-hop reasoning violates this assumption. Let $F_q(E)$ denote the probability that a fixed reader answers correctly

given evidence set E :

$$F_q(E) = P(\hat{a} = a^* \mid q, E). \quad (41)$$

The marginal utility of an evidence unit e under current context C is

$$\Delta_q(e \mid C) = F_q(C \cup \{e\}) - F_q(C). \quad (42)$$

A single-hop task is often close to monotone submodular selection: adding a relevant document helps and marginal gains diminish. Multi-hop tasks are frequently non-submodular because a bridge document may have near-zero utility alone but large utility after its companion evidence is present.

Definition 3 (Complementarity gap). For evidence units e_i, e_j and context C , define

$$\Gamma_q(e_i, e_j \mid C) = \Delta_q(e_j \mid C \cup \{e_i\}) - \Delta_q(e_j \mid C). \quad (43)$$

Positive Γ_q indicates complementarity: the value of e_j increases after e_i is known.

Proposition 3 (Why query-only relevance misses bridges). *Suppose a bridge evidence unit e_b has low marginal relevance $s_{\text{rel}}(q, e_b)$ but positive complementarity $\Gamma_q(e_c, e_b \mid C) > 0$ with a core unit e_c . Any selector that is monotone in $s_{\text{rel}}(q, e)$ alone can rank e_b below unrelated high-surface-match distractors even when $\Delta_q(e_b \mid C \cup \{e_c\})$ is high.*

PROOF. A query-only selector is a function of $s_{\text{rel}}(q, e)$ and does not condition on $C \cup \{e_c\}$. If there exists a distractor d such that $s_{\text{rel}}(q, d) > s_{\text{rel}}(q, e_b)$, the selector will prefer d regardless of the conditional gain of e_b . The inequality can hold even when $F_q(C \cup \{e_c, e_b\}) > F_q(C \cup \{e_c, d\})$, because that comparison depends on conditional utility rather than marginal surface relevance. The argument requires only that the selector's score function depends on (q, e) and not on C , and that the strict-inequality regime is non-empty—a condition that is empirically verified on bridge-heavy benchmarks such as MuSiQUE [167] and 2WikiMultiHopQA [55]. $\square$

This proposition is the formal reason why multi-hop reranking should not be evaluated only as relevance reranking. A selector may reduce average query-document relevance while improving chain preservation. Conversely, a strong cross-encoder reranker may improve top- K relevance while still dropping low-surface bridge evidence.

4.11 A Budgeted Chain-Preservation Objective

Given a pool L_N , context budget K , and cost function $c(e)$, an ideal selector solves

$$E_K^* = \arg \max_{E \subseteq L_N} \mathbb{E}_{C \sim p(C \mid q, L_N)} [\mathbb{I}\{C \subseteq E\}] \quad \text{s.t.} \quad |E| \leq K. \quad (44)$$

If evidence units have variable length, the cardinality budget becomes a token budget:

$$\sum_{e \in E} c(e) \leq B. \quad (45)$$

Compression methods can be interpreted as changing $c(e)$ and sometimes changing the evidence unit itself. A compressed note may preserve the answer-bearing fact but lose the bridge relation or the provenance needed for faithfulness. Therefore compression should be evaluated not only by answer accuracy but by chain preservation and citation validity [201, 220].

Lemma 1 (Upper bound under chain absence). *Let A be the event that no valid minimal chain is contained in the retrieved pool L_N . If the reader is required to answer only from the retrieved evidence and has no parametric leakage, then*

$$P(\hat{a} = a^* \mid A) \leq \epsilon, \quad (46)$$

where ϵ is the probability of guessing correctly from task priors or annotation artifacts.

PROOF. Under the no-leakage condition, the answer is identifiable only from a sufficient evidence chain. When no valid minimal chain is present, the reader lacks the information required to distinguish a^* from alternatives except through prior bias. The remaining success probability is therefore bounded by shortcut or guessing probability ϵ . Formally, let $\Pi(a \mid A)$ be the conditional answer distribution when no chain is present. Then for any reader output $\hat{a}$, $P(\hat{a} = a^* \mid A) = \Pi(a^* \mid A) \leq \max_a \Pi(a \mid A) \leq \epsilon$ by the assumption on prior coverage. $\square$

The lemma motivates conditional evaluation. If a method fails on examples where L_N lacks the chain, the failure should be attributed to observability, not to the reader. If a method succeeds in that regime, the success may indicate parametric memory or shortcut exploitation rather than retrieval-grounded reasoning.

4.12 Exposure as a Positional Treatment

Exposure can be treated as a causal treatment over a fixed evidence set. Let $Y(\pi)$ denote the answer-correctness outcome under ordering π . The average exposure effect between two ordering policies π_1 and π_0 is

$$\tau_{\text{exp}} = \mathbb{E} \left[Y(\pi_1) - Y(\pi_0) \mid C^* \subseteq E_K \right]. \quad (47)$$

This estimand is identifiable only if membership is held fixed. If the reranker changes both E_K and $\pi(E_K)$, gains conflate selection and presentation. This is why fixed-membership ordering experiments are not secondary ablations; they are required to isolate the exposure channel, especially on long contexts where positional bias is documented to be substantial [4, 97, 217].

4.13 Reader Fusion as Noisy Program Execution

A multi-hop reader can be idealized as executing a latent program over evidence units:

$$z_0 = q, \quad z_t = T_t(z_{t-1}, e_t), \quad \hat{a} = H(z_m), \quad (48)$$

where T_t may resolve an entity, project an attribute, compare values, or aggregate partial results. If each transition has error probability ϵ_t , then under a simple independent-error approximation,

$$P(\text{all transitions correct}) \approx \prod_{t=1}^m (1 - \epsilon_t). \quad (49)$$

This explains why longer chains are difficult even when every document is retrieved. It also explains why process supervision [95, 158, 198], program execution [11, 34, 101], external tools [142, 214], and step-level verification can improve multi-hop RAG without changing retrieval recall.

5 METHODS AS ESTIMATORS OF EVIDENCE-CHAIN QUANTITIES

This section reorganizes the method landscape by statistical estimand. Architectural families reappear, but as implementation choices rather than primary organizing categories. The subsections below deliberately cut across architecture. For example, graph expansion, iterative dense retrieval, and agentic search all appear under chain observability because they expand or adapt the set of candidate evidence units. Decomposition appears under fusion reliability when it reduces the complexity of composition, but it also appears under observability when sub-questions become retrieval queries. This overlap is not a flaw; it is the point of an estimand-based taxonomy. A method can be an estimator of more than one quantity, and a strong system often combines several estimators. Readers who prefer an architecture-first presentation should read this section together with Section 6, which catalogues the same methods under their conventional family labels.

5.1 Estimating Chain Observability

Methods in this group aim to increase $P(C^* \subseteq L_N)$. We organize them along increasing degrees of adaptivity: single-shot retrieval, iterative retrieval, graph expansion, and agentic search.

5.1.1 Single-shot Sparse and Dense Retrieval. Sparse retrievers such as BM25 [138] and tuned BM25 variants remain strong baselines for multi-hop QA when at least one hop is lexically anchored. Dense retrievers including DPR [73], REALM [48], ANCE [199], CONTRIEVER [59], GTR [119], E5 [181], BGE [195], and RETROMAE [194] provide scalable proposal pools by encoding question and passage into a shared dense space. Multi-vector and late-interaction retrievers such as ColBERT [75] and ColBERTv2 [139] preserve per-token matching signals that are useful when bridge entities appear only in narrow spans. Learned sparse retrievers such as SPLADE [31] combine inverted-index efficiency with neural expansion. The shared limitation of single-shot retrievers in the multi-hop setting is structural: if the second-hop bridge is not expressed in the original query, query-only retrieval may miss it, regardless of whether scoring is sparse, dense, or multi-vector.

5.1.2 Iterative and Reasoning-Interleaved Retrieval. Iterative retrieval directly addresses second-hop missingness. GOLDEN RETRIEVER [113] generates intermediate queries from previously retrieved passages. MDR [200] performs multi-step dense retrieval conditioned on accumulated evidence representations. BALEEN [74] maintains a condensed evidence state across hops to avoid query drift. MUPPET [28] pre-encodes potential bridge links and reuses them between hops. IRCOT [168] interleaves chain-of-thought reasoning with retrieval, treating each reasoning step as a new query. FLARE [69] triggers retrieval mid-generation whenever predicted-token probability falls below a threshold. ITER-RETGEN [147] alternates between generation-as-context-expansion and retrieval-as-evidence-refresh. ADAPTIVE-RAG [62] classifies query complexity to select between no-retrieval, single-step, and multi-step retrieval paths. DRAGIN [153] decides retrieval timing based on generator entropy and uncertainty signals. SELF-RAG [2] uses learned reflection tokens to decide both when to retrieve and which passages to keep. ITER-Dragon [191] composes iterative retrieval with stage-wise verifier feedback.

5.1.3 Graph Expansion as Observability Estimator. Graph-expansion methods can also be read as observability estimators. GRAFT-NET [155] and PULLNET [154] construct question-specific subgraphs by iteratively pulling neighboring entities and linked text. HGN [27] and DFGN [137] build hierarchical entity-and-paragraph graphs to surface bridge nodes that single-shot retrieval misses. ToG [156], ToG 2.0 [104], and RoG [100] traverse KG paths under LLM control, expanding the local search space until candidate paths become visible. KG-FiD [219] uses KG signals to prune and re-score retrieved passages. GNN-RAG [111] runs a GNN over a retrieved subgraph and feeds the resulting node representations to an LLM reasoner. GRAPH-RAG [24] constructs an LLM-built graph index over a corpus and uses community summaries for query-focused expansion. LIGHT-RAG [46] and HIPPO-RAG [47] provide alternative graph indexing strategies optimized for cost and memory-style retrieval.

5.1.4 Long-context and Hierarchical Retrieval. A different way to increase chain observability is to change the granularity of evidence units. Long-context readers can sometimes absorb the entire candidate pool, but lost-in-the-middle effects [97] prevent naive scaling. RAPTOR [140] builds recursive abstractive summaries of corpus clusters so that high-level bridge information becomes retrievable. LONG-RAG [68] pairs long-context readers with coarser retrieval units to reduce per-hop selection load. HiQA [14] introduces hierarchical context augmentation. HyDE [33] synthesizes hypothetical documents to anchor dense retrieval in semantic rather than lexical space, and is especially useful for queries whose surface form does not match relevant evidence.

5.1.5 Agentic and RL-trained Search Policies. Recent work moves observability estimation into a learned policy. REACT [214] interleaves reasoning and acting through tool calls including search. TOOLFORMER [142] learns when to invoke external tools through self-supervision. SEARCH-R1 [70] trains a reasoning model to generate search queries during step-by-step reasoning with reinforcement learning, with retrieval results masked out from policy gradients. R1-SEARCHER [152] incentivizes the search capability of LLMs through RL with intermediate retrieval rewards. DEEP-RAG [43] formulates retrieval-augmented reasoning as an MDP and decides at each step whether to retrieve or rely on parametric reasoning. RAG-GYM [198] adds process supervision for search agents, making intermediate search actions trainable. PAR-RAG [221] uses a plan-act-review structure to reduce reasoning-path drift. DUALRAG [16] couples reasoning-augmented querying with progressive knowledge aggregation. MULTICUBE-RAG [149] uses an ontology-guided cube structure to organize multi-hop retrieval without relying solely on noisy graph expansion. RESEARCH [9] learns to reason with search via reinforcement learning under sparse outcome rewards. O1-SEARCHER [179] and similar systems explore deliberate search in a long-thought regime. Their advantage is adaptive depth; their risk is query drift, redundant calls, and weak attribution of which action improved observability.

5.2 Estimating Selection Preservation

A chain can be present in L_N but absent from E_K because the selection model optimizes marginal relevance rather than conditional utility. Rerankers, bridge selectors, compression methods, and plan-guided selectors all aim to preserve the support chain under a small budget; their common internal mechanism is to approximate the conditional utility score $\theta_{\text{util}}(e \mid q, C)$, which serves as a per-candidate signal in service of the event-level estimand θ_{sel} .

5.2.1 Cross-encoder and LLM Rerankers. Classical rerankers [123] and stronger neural rerankers including MONOT5 [124], RANKT5 [239], RANKLLAMA [103], RANKGPT [157], RANKZEPHYR [132], and listwise LLM rerankers [163] score candidate documents by query relevance. In multi-hop settings, these scores are often misspecified: bridge documents can have low standalone relevance but high value after a partial chain has been observed. Some recent listwise LLM rerankers attempt joint scoring and are therefore closer to conditional utility estimation, but the gap from true conditional utility is rarely measured.

5.2.2 Bridge-aware and Core-conditioned Selection. Methods that condition selection on partial state begin to estimate true conditional utility. BALEEN's condensed retrieval [74] explicitly tracks accumulated evidence and reranks under that condition. HOPRETRIEVER [89] and PATHRETRIEVER [1] score candidate next-hop documents conditioned on entity or sentence state. BEAMDR [230] performs beam-search over chain hypotheses. MUGER2 [211] explicitly detects bridge entities. BRIDGER [203] adds a bridge-prediction objective to dense retrieval.

5.2.3 Compression and Note-based Methods. Compression can be interpreted as utility estimation under a token budget. LONGLLMLINGUA [65], and LONGLLMLINGUA-2 [128] compress prompts by predicting token importance. RECOMP [201] compresses retrieved passages into extractive or abstractive notes. CHAIN-OF-NOTE [220] writes intermediate notes that survive into the reader prompt. xRAG [15] compresses retrieved context into single tokens for efficient augmentation. The key statistical question for compression is whether chain-relevant complementary content survives, not whether the compressor preserves average relevance.

5.2.4 Plan-driven and Sub-question Selection. Sub-question and plan-based methods select evidence aligned with explicit reasoning steps. DECOMPRC [114], SELF-ASK [133], LEAST-TO-MOST [235], BREAK [190], PLAN-AND-SOLVE [180], HiPRAG [173], PROGRAMFC [126], and PAR-RAG's plan stage

all transform a complex question into sub-tasks whose matched evidence is more nearly single-hop. The estimand interpretation is that decomposition reduces the variance of utility estimation: a sub-question constrains conditional context C explicitly, so each candidate document can be scored under a clearer conditional.

5.2.5 Training Signals for Utility. Conditional utility can be supervised in several ways: gold supporting facts [167, 212]; oracle ablations producing leave-one-out labels; reader-derived utility scores; verifier scores [158, 205]; and counterfactual deletion [98]. Each signal has bias. Gold supports may not be unique. Reader-derived utility can inherit reader-specific preferences. Verifier scores may reward plausibility rather than causal support. A robust selector should ideally combine multiple signals while reporting which utility surrogate is used.

From this view, the central statistical issue is selection bias under budget. A top- K context is not a neutral sample of the retrieved pool; it is a biased estimator of the latent support chain. Future selection models should therefore report not only relevance metrics but also conditional chain preservation: whether the missing supports of C^* survive the transition from L_N to E_K .

5.3 Estimating Exposure Probability

Exposure methods assume the relevant evidence is selected but not necessarily usable. Long contexts and multi-document prompts create position-dependent access problems [97, 217].

5.3.1 Reader-aware Ordering and Packing. Reader-aware ordering places bridge and answer-bearing evidence earlier or closer together. LONGCHAT-AWARE packing strategies, “lost-in-the-middle” mitigations [97], LONGCONTEXTBENCH-driven layout heuristics [4], and explicit instruction templates that announce chain structure are examples of low-cost ordering techniques. Some methods learn the ordering: PRCA [210] trains a context-aware adapter that re-orders evidence based on reader gradients. POSITION-ROBUST CONTEXT [53] fine-tunes LLMs to be position-invariant.

5.3.2 Fusion Architecture as Exposure. Architectures that decouple encoding from fusion reduce encoder-side interference. FiD [60] separately encodes passages and fuses them in the decoder, making each passage’s contribution position-independent in the encoder. FiD-LIGHT [56] and KG-FiD [219] extend the same idea with efficiency or structural priors. ATLAS [61] co-trains retriever and FiD reader. Encoder-decoder separation is itself an exposure intervention because the reader no longer faces a single concatenated long prompt.

5.3.3 Compression-aware Exposure. Compression reduces exposure competition by shortening distractors. LLMLINGUA [65], RECOMP [201], and SELECTIVE-CONTEXT [91] all shrink the input and indirectly improve the reader’s effective access depth. In fixed membership experiments, compression can isolate the exposure channel because it changes neither S nor the underlying retrieval.

5.3.4 Fixed-Membership Ordering as a Causal Probe. A method can preserve evidence membership while still improving answer quality by changing presentation. Fixed-membership ordering ablations where only the order of E_K varies, holding $|E_K|$ and contents constant, isolate θ_{exp} . Section 11 recommends this experiment as a standard supplement.

5.4 Estimating Fusion Reliability

Fusion reliability asks whether the reader can compose the chain once it is selected and exposed.

5.4.1 Retrieve-then-Read Readers. Retrieve-then-read systems such as DRQA [8], ORQA [81], RAG [83], REALM [48], ATLAS [61], and FiD [60] are direct fusion estimators. They differ in

whether retrieval is fixed, jointly trained, or distilled, and in whether passages are concatenated or independently encoded. Their shared role is to estimate $P(\hat{a} \mid q, E)$.

5.4.2 Graph-aware Fusion. Graph-aware fusion methods such as QA-GNN [216], GREASELM [224], UNIKGQA [66], REAREV [110], SR [222], HAMQA [22], and NSM [52] provide structured inductive bias for cross-evidence composition. The graph constrains how evidence units can be combined and therefore reduces the hypothesis space the reader must search over.

5.4.3 Decomposition and Programmatic Reasoning. Decomposition-based methods factorize a difficult multi-hop query into easier subproblems. DECOMPRC [114], SELF-ASK [133], LEAST-TO-MOST [235], BREAK [190], PLAN-AND-SOLVE [180], DECOMPOSED PROMPTING [76], SUCCESSIVE PROMPTING [23], and REASONPATH [90] all reduce fusion complexity by making intermediate operations explicit. Programmatic methods such as PoT [11], PAL [34], FAITHFUL CoT [101], and CODE-AS-REASONING [185] make intermediate operations executable or symbolically checkable. The main fusion-side failure is error propagation: a wrong early sub-answer corrupts later retrieval and reasoning, an effect amplified by the $\prod (1 - \epsilon_i)$ form of equation in §4.13.

5.4.4 Self-Consistency and Diverse Reasoning. Sampling-based methods improve fusion through diversity. SELF-CONSISTENCY [186] marginalizes over multiple reasoning traces. TREE-OF-THOUGHT [213] and GRAPH-OF-THOUGHT [5] expand the reasoning topology beyond linear chains. ALGORITHM-OF-THOUGHT [145] introduces algorithmic templates. AUTO-CoT [229] automatically constructs in-context exemplars. These methods do not change retrieval but increase the probability that at least one trajectory composes the chain correctly.

5.4.5 Latent and Compressed Reasoning Inside Readers. A separate fusion thread compresses reasoning into hidden states. QUIET-STAR [40], COCONUT [50], pause-token methods [39], and “think before speak” variants generate internal computation that is not emitted as tokens. In the latent-chain view, these methods do not eliminate fusion error; they relocate it to model internals where it is harder to attribute. Empirically they can improve multi-hop accuracy on chain-explicit tasks [176] while making external diagnostics less informative.

5.5 Estimating Causal Faithfulness

Faithfulness methods estimate whether the answer is causally grounded in evidence rather than merely correct.

5.5.1 Self-verification. Self-verification methods use model-internal consistency or critique. SELF-REFINE [105], REFLEXION [150], CoVe [20], SELF-CHECKGPT [108], and SELF-VERIFICATION [189] score intermediate or final outputs by re-asking the model. They are cheap but bounded by the model’s own knowledge and biases.

5.5.2 Externally-grounded Verification. Externally-grounded methods retrieve or check additional evidence before revising outputs. CRITIC [38], RARR [32], VERIFY-AND-EDIT [232], CRAG [207], SELF-CHECKER [92], and ATTRIBUTABLE-TRACE [84] ground critique in retrieved content. SELF-RAG [2] combines retrieval and critique tokens in a single decoding loop.

5.5.3 Process Reward Models. Process-supervised reward models shift faithfulness from post-hoc evaluation to a training signal. LIGHTMAN PRM [95], MATH-SHEPHERD [184], REARTeR [158], RAG-GYM [198], and VERITAS [205] reward intermediate reasoning steps or evidence-grounded actions rather than only final correctness. RESt-MCTS [223] and similar systems combine search and process supervision.

5.5.4 Counterfactual and Deletion Tests. Counterfactual benchmarks such as CoFCA [192] test whether changing a bridge variable changes the answer. Deletion tests [98, 202] remove a candidate supporting passage and measure whether the answer changes. Both provide interventional proxies for causal faithfulness, but neither identifies θ_{faith} without additional assumptions.

5.5.5 Citation and Attribution Models. Citation-aware generators WEBGPT [116], GopherCITE [112], and ALCE [35] produce per-claim citations that can be validated for entailment. Citation precision is necessary but not sufficient for faithfulness, as a citation can be textually relevant without being causally used.

Under the estimand view, all of these methods target Δ_{faith} . They differ in whether they intervene pre-hoc, in-decoding, or post-hoc, and in whether the supervision signal is internal, external, or interventional. We return to causal faithfulness in detail in Section 9.

5.6 Joint Estimation in Agentic and Hybrid Systems

Hybrid and agentic systems estimate several quantities simultaneously. SELF-RAG [2] decides when to retrieve, critiques evidence, and generates. SEARCH-R1 [70] and DEEP-RAG [43] learn retrieval and reasoning policies with RL. LM-KG HYBRIDS such as ToG 2.0 [104], RoG [100], KAG [94], HIP-PO-RAG [47], LIGHT-RAG [46], SUBGRAPH-RAG [88], and GRAPH-RAG [24] combine structured chain priors with flexible language reasoning. Multi-agent systems such as MA-RAG [118], REAGENT [178], and CHAIN-OF-AGENTS [228] distribute decomposition, retrieval, reasoning, and verification across roles.

The advantage of joint estimation is end-to-end adaptability. The cost is attribution: when answer accuracy improves, it is hard to tell whether the gain came from observability, selection, exposure, fusion, or faithfulness. We therefore recommend that future agentic and hybrid systems report component-level diagnostics aligned with the five estimands rather than only final answer scores.

5.7 When Method Families Collide

Recent systems increasingly combine several families: a method may use a planner, a dense retriever, a graph index, a long-context reader, and a verifier. Architectural labels become less informative in such hybrids. The estimand view avoids this problem by decomposing a hybrid into the quantities it targets. For example, a graph-augmented agent may use graph neighbors to improve observability, an LLM planner to improve conditional utility, a context packer to improve exposure, and a verifier to improve faithfulness. The method should then be evaluated at each link rather than described only as “GraphRAG” or “agentic RAG.”

By Proposition 1, a strong retriever may not improve grounded accuracy when the reader cannot fuse evidence. A strong reader may not help when the chain is absent from the pool. A faithfulness verifier may reject unfaithful answers but not recover missing evidence. Therefore the most credible empirical papers are those that show where in the product the gain occurs.

5.8 Mechanism-Level Analyses as an Internal Estimator

The estimands θ_{obs} , θ_{sel} , θ_{exp} , θ_{fus} , and θ_{faith} are defined over externally observable quantities: retrieved pools, selected contexts, exposed positions, generated answers. A growing line of mechanistic-interpretability work asks a complementary question: does the chain object exist *inside* the model, and can we measure it there? DISSECTING MULTI-HOP [37] traces how intermediate facts are recalled and combined in transformer layers. DO LLMs LATENTLY PERFORM MULTI-HOP REASONING? [208] shows that some models recall bridge entities in middle layers but use them inconsistently downstream. HOPPING [6] characterizes step-by-step recall across layers. GROKED

Table 6. Recent 2025–2026 multi-hop RAG methods interpreted by primary estimand.

Method	Mechanism	Primary estimand	Statistical interpretation
Search-R1 [70]	RL-trained search-query generation	Observability / control	Learns a policy over retrieval actions that increases probability of useful external evidence.
DeepRAG [43]	MDP over retrieve-or-reason decisions	Cost-aware observability	Trades retrieval calls against expected information gain at each reasoning step.
R1-Searcher [152]	RL-incentivized search via reward shaping	Observability	Improves bridge discovery through intermediate retrieval reward.
ReSearch [9]	RL search under outcome-only reward	Observability / control	Demonstrates that pure RL can yield effective search agents.
RAG-Gym [198]	Process supervision for search agents	Observability and selection	Provides intermediate rewards for search actions rather than only final-answer reward.
PAR-RAG [221]	Plan-act-review with verification	Fusion and faithfulness	Controls reasoning drift by reviewing intermediate results.
DualRAG [16]	Reasoning-augmented querying plus knowledge aggregation	Selection / fusion	Couples query generation with progressive evidence integration.
VERITAS [205]	Faithfulness-aware process rewards	Causal faithfulness	Optimizes reasoning traces for entailment and support, not just correctness.
ReARTeR [158]	Trustworthy PRM for retrieval-augmented reasoning	Faithfulness / fusion	Process-level reward model trained for retrieval-grounded steps.
MultiCube-RAG [149]	Ontology-guided cube retrieval	Structural observability	Restricts retrieval to semantically organized dimensions to reduce graph noise.

TRANSFORMERS [176] demonstrate that generalization on multi-hop tasks emerges with extended training and specific circuit formation.

In the latent-chain view, these analyses provide a third source of information about C^* , in addition to retrieved evidence and external annotations. They can be read as *internal estimators* of two quantities: (i) whether C^* or a proxy of it is representable inside the reader (a necessary condition for high θ_{fus} under any external evidence), and (ii) whether the answer is produced by combining the intermediate representations of the chain or by parametric shortcut (an internal lower bound on θ_{faith}). Mechanistic analyses are not yet routinely integrated with retrieval-augmented systems, but they suggest a future diagnostic layer that combines external chain-preservation metrics with internal circuit-level evidence use.

6 ARCHITECTURAL FAMILIES: A COMPLEMENTARY REVIEW

The estimand-based taxonomy in Section 5 does not replace architectural taxonomies; it explains why they are incomplete. To preserve continuity with prior surveys, this section provides a compact architectural crosswalk. Rather than re-listing methods already discussed in Section 5, we give each family one short paragraph that (i) names its defining architectural commitment, (ii) points to where its representative methods are already reviewed by estimand, and (iii) flags substrates

Graph-based methods as chain-structure estimators

Graphs improve observability through expansion and fusion through path constraints.

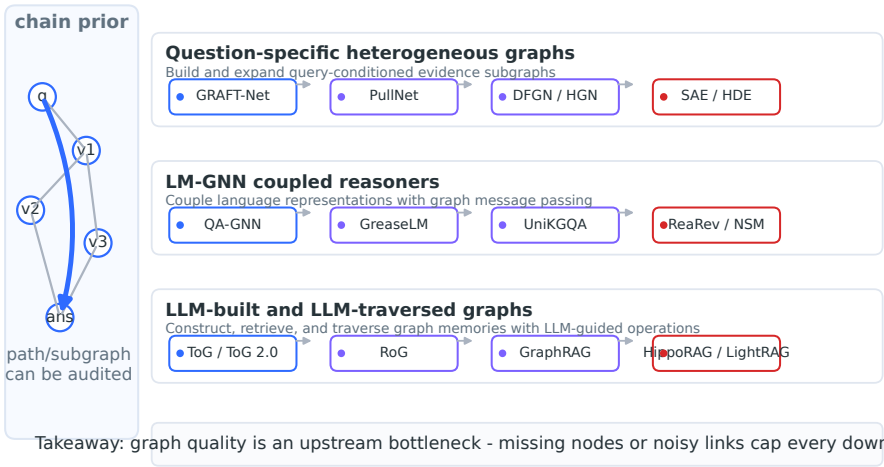


Fig. 3. Graph-based methods as chain-structure estimators. Graph methods improve chain observability by expanding local neighborhoods, constrain fusion through structured paths, and improve faithfulness when paths or subgraphs can be audited.

Table 7. Architectural families mapped to latent-chain estimands. A checkmark indicates a primary contribution; (o) indicates a secondary contribution. Methods are reviewed by estimand in Section 5; this section provides the architectural cross-reference only.

Family	Observability	Selection	Exposure	Fusion	Faithfulness
Retrieval-based	✓	(o)			
Graph-based	✓	(o)		✓	✓
Decomposition-based	(o)	(o)		✓	(o)
Fusion-based readers			(o)	✓	
LLM-based reasoning	(o)	(o)	(o)	✓	(o)
Verification-based				(o)	✓
Long-context / hierarchical	(o)	(o)	✓	(o)	
Hybrid / agentic	✓	✓	(o)	✓	✓

or lineages that fall outside the estimand-first exposition. Table 7 summarizes the mapping, while Figure 3 isolates the main graph-method lineages and the chain properties they target.

Retrieval methods by adaptivity and retrieval-state richness

The map describes control and state representation; it does not assert a performance ordering.

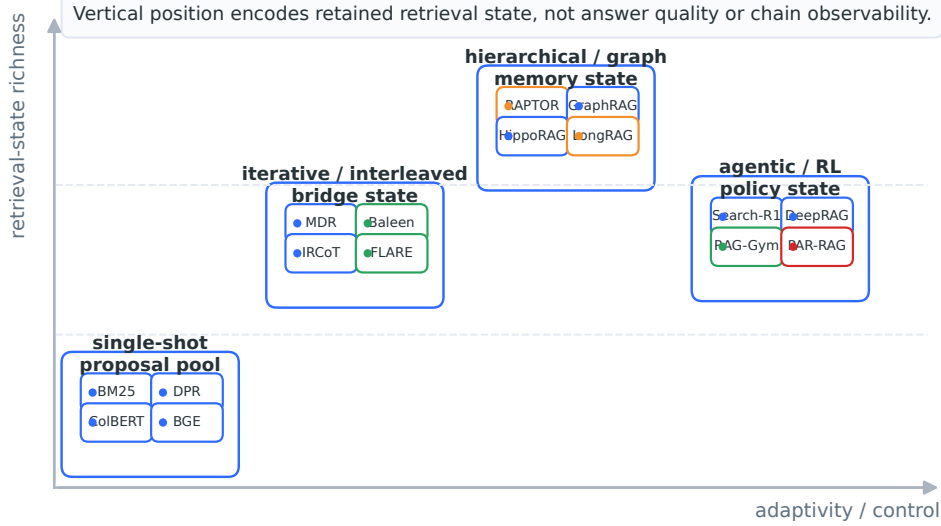


Fig. 4. Retrieval-based methods mapped by adaptivity (horizontal) and retrieval-state richness (vertical). Neither axis encodes answer quality or chain observability; the boxes locate mechanism families and list representative systems.

6.1 Retrieval-based Methods

Figure 4 organizes retrieval control by the degree of adaptivity granted to the evidence-acquisition policy. Retrieval-based systems commit to enlarging or reshaping L_N as the primary lever. Single-shot sparse and dense retrievers, iterative and reasoning-interleaved retrievers, adaptive controllers, hierarchical and long-context retrievers, and corrective-retrieval variants are all reviewed as observability estimators in Section 5.1. The architectural commitment of this family is that better proposal-pool construction is worth the added cost and drift; its estimand contribution is θ_{obs} , with θ_{sel} additionally engaged when the retriever also reranks conditioned on partial state (where conditional utility is used as the internal mechanism score driving selection preservation).

6.2 Graph-based Methods

Graph-based methods commit to structured fusion: paths are first-class and can be audited hop by hop. Question-specific entity or document graphs (GRAFT-NET, PULLNET, DFGN, HGN), LM-GNN couplings (QA-GNN, GREASELM, UNIKGQA, REAREV, NSM, KG-FiD), and LLM-built or LLM-traversed graphs (ToG, ToG 2.0, RoG, GRAPHRAG, LIGHTRAG, HIPPO-RAG, SUBGRAPH-RAG, KAG) are reviewed as observability, fusion, or faithfulness estimators in Sections 5.1, 5.4, and 5.5. Additional works that mostly matter for the graph-construction step rather than for downstream reasoning include SAE [170], HDE [171], EMBEDKGQA [141], TRANSFERNET [148], and KGP [187]. The family's shared limitation is that construction quality (missing nodes, noisy entity linking, incomplete edges) reduces observability before reasoning is invoked; LLM-built graphs shift the bottleneck to extraction reliability.

6.3 Decomposition-, Fusion-, LLM-Reasoning-, and Verification-based Families

Four architectural families overlap heavily with Section 5 and therefore need only a short pointer here. *Decomposition-based* methods (sub-question, programmatic, and plan-driven variants) are reviewed as utility and fusion estimators in Sections 5.2 and 5.4; their bottleneck is cascading error and granularity tuning. *Fusion-based readers* (retrieve-then-read backbones, compression-aware fusion, long-context and memory-augmented readers) are downstream estimators of θ_{fus} under θ_{exp} ; they cannot repair missing evidence and are reviewed in Section 5.4. *LLM-based reasoning* spans linear chain-of-thought, non-linear topologies (ToT, GoT, AoT, SoT [120], CUMULATIVE REASONING [226]), programmatic reasoning, latent reasoning, tool-use and agentic reasoning (ToolLLM [136], GORILLA [130], VOYAGER [177], GENERATIVE AGENTS [129]), and search-based reasoning (LATS [234], REASONING-VIA-PLANNING [51]); each variant is placed under its primary estimand in Sections 5.1 through 5.5. *Verification-based* methods are structurally identical to the causal-faithfulness estimators of Section 5.5 (self-verification, externally grounded verification, process reward models, counterfactual and deletion tests); the architectural label adds no independent estimand contribution.

6.4 Long-context, Multimodal, and Semi-structured Substrates

Two substrate directions deserve their own architectural note because their evidence spaces differ from standard text corpora. *Long-context reasoning* is stressed by LONGBENCH [4], ∞ BENCH [225], and NOVELHOPQA [125], motivated by lost-in-the-middle effects [97] and enabled by LONGLLAMA [172] and MAMBA [41]; retrieval-side adaptations include LONGRAG [68] and RAPTOR [140]. *Multimodal and tabular multi-hop* requires alignment across text, tables, and images: MULTIMODALQA [162], WEBQA [7], MANYMODALQA [49], MMRA [82], VisDoM [183], DocHop-QA [159], WIKIMIXQA [54], IMG2LLM [45], LLaVA-RAG [96], HYBRIDQA [13], OTT-QA [10], TABFACT [12], FeTaQA [117], and TAT-QA [238]. These substrates change what counts as an evidence unit e but leave the five estimands intact; their main statistical challenge is that both θ_{obs} and θ_{sel} become modality-conditional and typically decrease under cross-modal composition.

6.5 Hybrid and Agentic RAG

Hybrid and agentic systems combine several families into a single policy: SELF-RAG, SEARCH-R1, R1-SEARCHER, RESEARCH, DEEP-RAG, MA-RAG, REAGENT, CHAIN-OF-AGENTS, and the KG-integrated variants (KAG, HIPPO-RAG, GRAPH-RAG, LIGHT-RAG, SUBGRAPH-RAG) are reviewed as joint estimators in Section 5.6. Their architectural distinctness lies not in a new estimand but in *attribution difficulty*: end-to-end gains are entangled across all five events. By Proposition 1, any headline improvement in grounded end-to-end success must be traceable to an improvement in at least one of the five component events. Without per-factor diagnostics, such a gain cannot be distinguished from a shortcut-driven improvement in the non-grounded term of equation 31; this section therefore treats hybrid/agentic RAG as a family that most urgently needs the estimand-level reporting recommended in Section 11.

7 BENCHMARKS AS DIAGNOSTIC INSTRUMENTS

Figure 5 gives the benchmark-level crosswalk used throughout this section. Benchmarks differ less by domain than by which latent event they make observable. We organize benchmarks by diagnostic function rather than by dataset family. A useful benchmark, under the latent-chain view, exposes at least one of the five estimands while controlling for the others.

Benchmark-estimand alignment matrix

Benchmarks are diagnostic instruments; they do not all test the same latent event.

	Observability	Selection preservation	Exposure	Fusion	Faithfulness
Supporting-fact QA	high	med	low	low	low
HotpotQA, 2Wiki, IIRC	high	med	low	low	low
Shortcut-resistant QA	high	med	low	low	low
MuSiQue, MoreHopQA	low	med	low	high	med
Path / process QA	low	med	low	high	low
EntailmentBank, eQASC	low	low	high	med	low
Heterogeneous evidence	low	low	low	med	high
HybridQA, MultiModalQA	low	low	low	med	high
Long-context multi-hop	low	low	low	med	high
LongBench, NovelHopQA	low	low	low	med	high
Fact verification	low	low	low	med	high
FEVER, HoVer, SciFact	low	low	low	med	high
RAG-specific diagnostics	low	low	low	med	high
FRAMES, MultiHop-RAG	low	low	low	med	high
Counterfactual QA	low	low	low	med	high
CofCA variants	low	low	low	med	high
Agentic / tool-use	low	low	low	med	high
ToolBench, WebArena	low	low	low	med	high

Strong empirical coverage should include at least one observability/shortcut benchmark and one faithfulness or counterfactual diagnostic.

Fig. 5. Benchmark–estimand alignment. Benchmark families expose different latent-chain estimands, including observability, selection preservation, exposure, fusion, and faithfulness.

7.1 Coverage and Chain-Observability Diagnostics

HOTPOTQA [212], 2WIKIMULTIHOPQA [55], IIRC [30], COMPLEXWEBQUESTIONS [161], QANGAROO [188], and open-domain variants of multi-hop QA primarily test whether systems can retrieve enough evidence to make the latent chain observable. Supporting-fact labels provide partial observability of $\mathcal{P}$ and $\mathcal{S}$, although they do not always identify the chain actually used by the model. The classic weakness of HOTPOTQA is that many examples can be answered correctly from a single paragraph due to lexical artifacts; the “distractor setting” provides better selection diagnostics than the “full-wiki setting,” but neither fully isolates membership from ordering effects.

7.2 Shortcut-Resistance Diagnostics

MUSIQUE [167] was designed by composing single-hop questions and filtering shortcuts, making it a stronger test of true chain observability than datasets where one paragraph may suffice. MOREHOPQA [144] stresses longer chains, where observability compounds multiplicatively. NOVELHOPQA [125] uses long-form narrative sources where chain reasoning interacts with long-context layout. HOTPOTQA-NO SHORCUT [166] provides a filtered subset that requires real multi-hop. These benchmarks are valuable because they reduce the chance that high answer accuracy comes from shallow cues rather than successful chain inference.

7.3 Path and Process Diagnostics

2WIKIMULTIHOPQA supplies explicit reasoning paths. ENTAILMENTBANK [18] provides entailment trees over scientific facts. EQASC [63] provides explanation chains over commonsense reasoning. PROOFWRITER [160] provides synthetic proofs. HOVER [67], EX-FEVER [102], and

FOOLMETWICE [25] annotate multi-step evidence for verification. These benchmarks help diagnose $\mathcal{U}$, because they make some intermediate structure visible rather than evaluating only final answers.

7.4 Heterogeneous-Evidence Diagnostics

CWQ [161], HYBRIDQA [13], OTT-QA [10], MULTIMODALQA [162], WEBQA [7], MANY-MODALQA [49], TABFACT [12], TAT-QA [238], FeTAQA [117], VisDoM [183], DocHop-QA [159], and WIKIMIXQA [54] stress fusion across structured and unstructured sources. Their diagnostic value lies less in raw answer accuracy than in whether a method can maintain the chain across different evidence types.

7.5 Counterfactual and Faithfulness Diagnostics

COFCA [192], citation-faithfulness benchmarks ALCE [35], HALUEVAL [87], EXPERTQA [106], and fact-verification datasets such as FEVER [165], HoVER [67], EX-FEVER [102], SciFACT [174], and AVERITeC [143] probe whether the answer changes when bridge variables or supporting facts are perturbed. These benchmarks are closest to measuring θ_{faith} , because they ask whether evidence is causally load-bearing rather than merely cited.

7.6 Long-context Multi-hop Diagnostics

LongBENCH [4], ∞ BENCH [225], HELMET [218], NOVELHOPQA [125], NARRATIVEQA [77], LOONG [182], and ZEROSCROLLS [146] stress chain construction under long context, where exposure failures dominate over selection failures.

7.7 RAG-specific and Agentic Diagnostics

MULTIHOP-RAG [164], FRAMES [78], CRAG [209], FANOUTQA [237], and RAGTRUTH [121] evaluate end-to-end RAG behavior under multi-step queries. TOOLBENCH [135] and AGENTBENCH [99] test agentic search behavior. WEBARENA [236] introduces realistic agent-environment interaction.

7.8 Metric Alignment

A metric should be matched to the estimand it claims to measure. Answer EM and F1 are deployment-level metrics, not mechanism-level metrics. Supporting-fact F1 approximates evidence membership but does not establish causal use. Retrieval recall at Top-N estimates pool observability but says little about the final reader context. Citation precision estimates attribution correctness but can be high even when the answer was produced from parametric memory. Counterfactual sensitivity is closer to causal faithfulness but depends on the quality of the intervention.

7.9 Dataset Families Reinterpreted

Traditional benchmark descriptions emphasize source domain and release year. The estimand view emphasizes which latent event a benchmark makes measurable. HOTPOTQA is valuable because sentence-level support labels enable pool and context diagnostics, but its shortcut artifacts mean that answer accuracy can overstate chain use. 2WIKIMULTIHOPQA adds explicit path structure, making it useful for path-level observability and fusion analysis. MUsIQUE is valuable because it reduces single-hop shortcuts and stresses genuine bridge observability. IIRC makes missing context explicit, placing retrieval actions at the center. FEVER, HoVER, and EX-FEVER foreground support validity rather than answer-string matching. HYBRIDQA and MULTIMODALQA stress fusion because a correct chain may cross tables, linked passages, and other semi-structured sources.

This reinterpretation changes how benchmark coverage should be reported. A paper that evaluates only on shortcut-prone datasets primarily shows that the system can exploit available signals; it does not establish robust latent-chain inference. A paper that evaluates only on verification datasets

Table 8. Representative benchmark families and the latent event they most directly expose.

Benchmark family	Examples	Primary estimand	Main caution
Supporting-fact QA	HotpotQA, 2WikiMultiHopQA, IIRC, ComplexWebQuestions	Observability and selection	Shortcut artifacts may inflate answer accuracy.
Composed / shortcut-resistant QA	MuSiQue, MoreHopQA	Bridge observability	Lower scores may reflect pool absence rather than reader weakness.
Path-annotated QA	2WikiMultiHopQA, EntailmentBank, eQASC, ProofWriter	Path validity / fusion	Gold path may not be unique.
Heterogeneous evidence	HybridQA, Multi-ModalQA, OTT-QA, WebQA, TAT-QA, FeTaQA, VisDoM	Fusion reliability	Alignment errors can dominate retrieval errors.
Long-context multi-hop	LongBench, ∞ Bench, HELMET, NovelHopQA	Exposure	Membership and order are not separately controlled by default.
Fact verification	FEVER, HoVer, EX-FEVER, SciFact, AVeriTeC	Faithfulness and support validity	Veracity labels do not always identify generation causality.
RAG-specific diagnostics	MultiHop-RAG, FRAMES, FanOutQA, RAGTruth, CRAG-bench	End-to-end RAG behavior	Requires separating retrieval, selection, citation, and answer quality.
Counterfactual QA	CofCA and related perturbation sets	Causal faithfulness	Expensive annotation and possible unnatural perturbations.
Agentic / tool use	ToolBench, AgentBench, WebArena	Joint observability and control	Difficulty depends on tool affordances.

may test faithfulness without testing open-domain chain observability. A strong empirical section should therefore include at least one benchmark for observability, one for shortcut resistance, and one for faithfulness or counterfactual sensitivity.

8 APPLICATIONS

Applications differ by which estimand dominates deployment risk. Rather than repeat the estimand vocabulary for each domain, we consolidate seven representative deployment settings into a single crosswalk (Table 10): each row identifies the dominant estimand, the secondary estimand that most often determines whether a deployment succeeds once the dominant one is addressed, the empirical instruments (benchmarks and representative systems) currently used in that setting, and the failure mode that practitioners should watch for. The crosswalk is intended as an operational quick reference; the vocabulary itself is defined in Section 4 and does not change across domains.

Two entries in the crosswalk deserve a note that does not fit in a row. In *enterprise RAG*, the traceability requirement is regulatory rather than statistical: even when θ_{sel} and θ_{faith} are individually adequate, deployments still require per-answer citation surfaces that survive compliance review, which is why graph-indexed pipelines dominate the space [24, 46]. In *scientific literature assistance*, the fusion challenge is non-standard because evidence units differ not only in surface form (text, table, figure) but in epistemic status (claim, method, result), and a chain that composes a claim from one paper with a result from another can be internally coherent yet scientifically invalid; citation-

Table 9. Metric–estimand alignment. A strong paper should avoid using a metric from one row as evidence for a claim in another row without an explicit caveat.

Claim	Suitable metric	Conditional subset	Misleading shortcut
Pool contains the chain	support/path recall at Top- N	all examples or by hop count	Answer F1 alone.
Budget preserves the chain	selected support/path recall at K	chain-present in pool	Reranker relevance score.
Ordering helps the reader	early support exposure; fixed-set order ablation	selected-chain-present	Changing membership and order together.
Reader can compose evidence	oracle-evidence EM/F1; step accuracy	oracle or selected-chain-present	Top- K retrieval recall.
Answer is faithful	deletion sensitivity; counterfactual support tests; citation entailment	answer-correct subset	Citation overlap alone.
System is efficient	latency, token cost, retrieval calls, verifier calls	same answer-quality band	Selector-only latency without reader cost.

and claim-level counterfactual checks are the current best proxy for this hazard. Both cases are consistent with the estimand vocabulary — they simply move the binding constraint outside the numerical decomposition.

9 FAITHFULNESS AS CAUSAL EVIDENCE USE

We argued in Section 4.5 that faithfulness is a causal estimand. A high-quality multi-hop RAG system should produce answers that are causally supported by the retrieved chain rather than answers that merely co-occur with plausible-looking citations. This section develops the causal view in detail, connects it to existing faithfulness diagnostics, and explains why several common metrics are necessary but not sufficient. We also use the running example (Section 1.1) to illustrate each failure mode.

9.1 A Causal Graph for Multi-Hop RAG

Let W denote the world state, $\mathcal{K}$ the knowledge store derived from W , $C^\star$ the latent support chain, L_N the retrieved pool, E_K the selected context, $\pi(E_K)$ the ordering, and $\hat{a}$ the answer. A simple causal graph for a faithful system is

$$W \rightarrow \mathcal{K} \rightarrow C^\star \rightarrow L_N \rightarrow E_K \rightarrow \pi(E_K) \rightarrow \hat{a}, \quad (50)$$

with the parametric memory of the reader Θ acting as an exogenous influence on $\hat{a}$:

$$\Theta \rightarrow \hat{a}. \quad (51)$$

A causally faithful answer is one whose distribution responds to interventions on $C^\star$ through $\pi(E_K)$, i.e.

$$P(\hat{a} \mid \text{do}(\pi(E_K) = \pi')) \neq P(\hat{a} \mid \pi(E_K)) \quad \text{whenever} \quad P(a^\star \mid \pi(E_K)) \neq P(a^\star \mid \pi'). \quad (52)$$

An unfaithful answer can be partitioned into two cases: *parametric leakage*, in which $\hat{a}$ is generated through Θ alone and is invariant under context interventions; and *shortcut use*, in which $\hat{a}$ depends on a non-evidence feature of $\pi(E_K)$ such as template wording, length, or surface lexical overlap.

Table 10. Applications organised by which latent estimand dominates deployment risk. “Dominant” is the estimand that most often bounds end-to-end grounded success in the domain; “Secondary” is the next binding factor once the dominant one is addressed. Benchmarks and representative systems are illustrative rather than exhaustive; see Section 7 for full benchmark diagnostics and Section 5 for the underlying method families.

Domain	Dominant estimand	Secondary estimand	Representative benchmarks / systems	Characteristic risk
Open-domain multi-hop QA	θ_{obs}	θ_{sel}	HOTPOTQA, MuSiQUE, 2WIKIMULTIHOPQA; SELF-RAG [2], ADAPTIVE-RAG [62], SEARCH-R1 [70], DEEP-RAG [43]	Depth–cost tradeoff: iterative retrieval improves observability but inflates noise and latency.
Knowledge-graph QA	θ_{obs}	θ_{faith}	KAG [94], GRAPH-RAG [24], HIPPO-RAG [47], ToG 2.0 [104], RoG [100]	Coverage is bounded by graph completeness; path-level auditability is the deployment argument.
Fact verification	θ_{faith}	θ_{fus}	FEVER [165], HoVER [67], EX-FEVER [102], SciFACT [174], AVERITeC [143], PROGRAMFC [126], MULTIFC [3]	Output is a label but the latent-chain problem is unchanged; unsupported “correct” verdicts are the primary hazard.
Dialogue & interactive systems	θ_{obs}	Action controllability	WIZARD OF WIKIPEDIA [21], DIALDOC [29], CHAT-RAG [204]	Missing link may sit in history, in the corpus, or in the user; the retrieval-vs-ask decision drives quality.
Legal, medical, scientific reasoning	θ_{faith}	θ_{fus}	LEGALBENCH [44], MEDQA [71], PUBMEDQA [72], BIOASQ [169], MIRAGE [197], QASPER [19], SciQAG [175]	Heterogeneous evidence (statutes, records, tables, figures); explicit chains and counterfactual verification are load-bearing.
Enterprise & customer-support RAG	θ_{sel}	θ_{faith}	GRAPH-RAG [24], LIGHT-RAG [46], internal-graph deployments	Short policy clauses become decisive only after user constraints are identified; traceability requirement is external (compliance).
Scientific literature assistance	θ_{fus}	θ_{faith}	PAPERQA [79], SCISUMMNET-style multi-hop summarisation [215], HiQA [14]	Cross-paper terminology drift and assumption comparison; correct-sounding synthesis without exact support is the deployment risk.

9.2 Deletion and Counterfactual Sensitivity

Two related interventions probe causal use.

Deletion sensitivity. For each selected evidence unit $e \in E_K$, define

$$S_{\text{del}}(e) = P(\hat{a} = a^* \mid \pi(E_K)) - P(\hat{a} = a^* \mid \pi(E_K \setminus \{e\})). \quad (53)$$

A faithful answer should exhibit a positive deletion sensitivity on support units of the latent chain. SUFFICIENT EVIDENCE metrics [98] and the ATTRIBUTABLE-TRACE framework [84] implement variants of this idea.

Counterfactual bridge sensitivity. Let e_b^* be a bridge unit whose factual content can be altered to obtain a counterfactual version e_b' encoding a different but consistent alternative (e.g., replacing the bridge identity “Shirley Temple” with another cast member and updating the downstream biography evidence consistently). Counterfactual benchmarks [192] test whether the answer shifts accordingly. A reader that answers correctly regardless of e_b' is exploiting parametric memory; a reader whose answer shifts to remain consistent with the counterfactual bridge is using the chain causally.

9.3 Correctness, Citation, and Faithfulness Are Distinct

A common methodological mistake is to treat citation correctness and answer correctness as faithfulness. They are necessary signals but not sufficient.

Remark 2 (Three independent failure modes). For each example, the joint outcome ($\hat{a} \stackrel{?}{=} a^*$, citations valid?, $\hat{a} \stackrel{?}{\leftarrow} C^*$) takes one of eight truth-value combinations. Six of the eight are pathological. Of particular interest are: correct answer with valid citation but no causal use (parametric leakage); correct answer without valid citation (citation suppression); valid citation with wrong answer (citation without composition). A single number such as answer F1 or citation precision cannot distinguish these cases.

A practical recommendation follows: faithfulness audits should report all three signals jointly, or at minimum stratify answer correctness by citation validity and by deletion sensitivity. Papers that report only one of the three numbers should be read with this caveat in mind.

9.4 Shortcut Exploitation and Parametric Leakage

Shortcut and leakage failures are not equally easy to detect. Shortcut exploitation usually leaves a measurable trace: removing a shortcut feature (such as the surface form of an answer entity in the question) changes accuracy. MUsIQUE [167] and shortcut-filtered subsets of HOTPOTQA [166] were constructed precisely to reduce this bias.

Parametric leakage is harder to detect because the leaked answer is plausibly produced by the chain. Closed-book and oracle-evidence ablations are useful but incomplete: closed-book accuracy estimates parametric availability, but a model can still leak even when oracle evidence is provided. Counterfactual benchmarks are the closest approximation, because they make leakage observable through the divergence between counterfactual chain and observed answer.

9.5 From Verification to Training Signal

Faithfulness has historically been treated as a post-hoc evaluation. Recent work moves it upstream into training. MATH-SHEPHERD [184] and the process reward model of LIGHTMAN ET AL. [95] reward intermediate reasoning steps rather than only final answers. REARTER [158] trains a trustworthy process reward model tailored to retrieval-augmented reasoning. VERITAS [205] uses faithfulness-aware reward signals. RAG-GYM [198] adds process supervision to search agents. PAR-RAG [221] inserts a review stage that catches drift before it propagates.

From the latent-chain view, all of these methods convert faithfulness from an evaluation property into an objective. They differ in whether the reward is supplied by humans, by a verifier, or by self-critique; in whether the reward is per-step or per-trajectory; and in whether the training data

Table 11. Failure events paired with diagnostics and recommended interventions.

Failure event	Diagnostic	Recommended intervention
Evidence absence	Pool support recall conditional on hop count	Iterative or graph-expansion retrieval; agentic search; coarser or hierarchical retrieval units.
Budgeted selection failure	Selected support recall under fixed K	Bridge-aware or core-conditioned selection; conditional-utility reranking.
Exposure failure	Order-only ablation on fixed E_K	Reader-aware ordering; compression; decoder-side fusion (FiD).
Fusion failure	Oracle-evidence reader accuracy	Decomposition; programmatic reasoning; structured graph fusion.
Faithfulness failure	Deletion and counterfactual sensitivity	Process reward; citation-aware decoding; counterfactual training.

includes interventional examples. Combining process supervision with counterfactual evaluation appears to be one of the most promising directions for measurable faithfulness gains.

9.6 Failure Modes Revisited

The five failure events in Section 3.3—evidence absence, budgeted selection failure, exposure failure, fusion failure, faithfulness failure—can now be paired with specific diagnostics.

10 STATISTICAL TRADEOFFS AND OPEN PROBLEMS

The latent-chain view reorganizes open problems as tradeoffs among estimands. Each tradeoff is a constraint: improving one estimand often worsens another. A successful method does not eliminate the tradeoff; it moves the system to a better point on the frontier.

10.1 Eight Open Problems

OP1. Joint optimization of observability and noise. Iterative and agentic retrieval increase θ_{obs} but inflate ρ_N . No widely accepted method jointly minimizes expected risk under a budget. Open directions include retrieval policies trained against joint chain-coverage and noise-ratio objectives, and verifier-in-the-loop pruning that operates on the pool rather than on the reader's output.

OP2. Conditional utility estimation under chain dependence. Most rerankers estimate marginal relevance. Conditional utility estimation under partial chains is still rare and is hard to supervise. Open directions include core-conditioned scoring, bridge-classification objectives, and synthetic counterfactual chains for utility regression training.

OP3. Reader-aware exposure under long context. Long-context models have not eliminated positional bias [97, 217]. Open directions include fixed-membership ordering policies, decoder-side fusion (FiD variants), context compression, and explicit attention scaffolds.

OP4. Auditable latent reasoning. Latent reasoning methods [40, 50] trade chain visibility for inference efficiency. Faithfulness audits based on emitted tokens cease to apply. Open directions include mechanistic probes for internal bridge representations and hybrid latent-token reasoning where the chain is occasionally externalized.

OP5. Causal evaluation for retrieval-augmented systems. Counterfactual evaluation is expensive. Most benchmarks still use factual data only. Open directions include synthetic counterfactual chain

Table 12. Multi-hop RAG tradeoffs framed as estimand tensions.

Tension	Estimands	Typical methods	What an honest evaluation should show
Coverage vs. noise	$\theta_{\text{obs}} \uparrow$ vs. $\rho_N \uparrow$	Iterative / agentic retrieval	Both pool recall and pool noise on the same examples.
Budget vs. chain preservation	$K \downarrow$ vs. $\theta_{\text{sel}} \uparrow$	Reranking, compression, bridge selection	Selected support recall under fixed K .
Exposure vs. budget	$\theta_{\text{exp}} \uparrow$ vs. $K \uparrow$	Reordering, packing, FiD	Fixed-membership ordering ablation.
Fusion power vs. latency	$\theta_{\text{fus}} \uparrow$ vs. cost $\uparrow$	Decomposition, programmatic reasoning, agents	Latency and tokens per question at fixed accuracy.
Plausibility vs. causal use	accuracy $\uparrow$ vs. $\Delta_{\text{faith}} \downarrow$	Verification, counterfactual training, PRM	Deletion and counterfactual sensitivity.
Controllability vs. adaptivity	step-level audit $\uparrow$ vs. policy freedom $\uparrow$	Multi-agent vs. end-to-end RL	Per-step diagnostics on the same model.

generation, deletion-based attribution at the chain level rather than at the token level, and joint citation-causal-use metrics.

OP6. Knowledge update without catastrophic faithfulness loss. Retrieval is a natural mechanism for keeping knowledge current, but edits in $\mathcal{K}$ can conflict with parametric memory. Open directions include retrieval policies that detect knowledge-update events, and reader objectives that prefer retrieved evidence over parametric memory under conflict [107, 196].

OP7. Cost-aware multi-hop RAG. Agentic and iterative methods increase latency and token cost. A deployable system must trade these against expected chain observability. Open directions include cost-sensitive policy learning, calibrated retrieval termination, and small-model verifiers paired with strong readers.

OP8. Safety and robustness under retrieval-side adversaries. Retrieval-side prompt injection and corpus poisoning can compromise otherwise faithful systems [206, 233, 240]. Open directions include trust-aware retrieval, multi-source agreement, and adversarial training for retrievers.

10.2 Tradeoffs Between Estimands

Because the five estimands share upstream design choices—retrieval depth N , context budget K , ordering π , reasoning depth, and verifier strength—improving one component event often changes another, sometimes in the opposite direction. Table 12 summarizes the six most common tensions we observe in the reviewed literature. Each row names the pair of quantities in tension, the method families that typically operate on that pair, and what an honest evaluation should report so that a headline gain in one factor is not silently paid for by regression in another.

10.3 Estimator Suggestions for Future Work

The estimands defined in Section 4 are not yet paired with widely adopted, comparable estimators. Current practice substitutes convenient surrogates—pool support recall for θ_{obs} , marginal relevance for θ_{sel} , average support position for θ_{exp} , oracle-evidence EM/F1 for θ_{fus} , and citation precision for θ_{faith} —each of which conflates the target estimand with unrelated confounds. Table 13 lists, for each estimand, the closest surrogate in the current literature and a concrete direction in which a

Table 13. Estimands, the closest existing surrogate metrics, and example estimator improvements suggested by the latent-chain view.

Estimand	Closest surrogate to-day	Improvement direction
θ_{obs}	Pool support recall at Top- N	Chain recall conditional on bridge gold; recall under controlled distractor density; agentic-action attribution.
θ_{sel}	Marginal relevance scores	Core-conditioned utility as internal mechanism score; bridge prediction; oracle leave-one-out under partial chains.
θ_{exp}	Average position of support	Fixed-membership ordering EM gap; access-depth-conditioned accuracy; lost-in-middle test sets.
θ_{fus}	Oracle-evidence EM/F1	Step accuracy under explicit chains; program execution validity; intervention robustness.
θ_{faith}	Citation precision	Joint citation + deletion + counterfactual + parametric-leakage decomposition.

more faithful estimator could be built. We do not claim any of these directions is solved; they are open items in Section 10 and are intended to focus follow-up work on the specific latent event each estimator is trying to identify.

11 REPORTING GUIDELINES FOR MULTI-HOP RAG

The latent-chain view directly implies a small set of reporting guidelines. We collect them here as a checklist for papers and reviewers.

11.1 Separate Pool, Context, and Reader Claims

A paper introducing a new multi-hop RAG system should explicitly distinguish three claims: a claim about the retrieved pool L_N , a claim about the budgeted context E_K , and a claim about the reader. Reporting only end-to-end accuracy can hide compensating effects. Concretely, we recommend at least one number from each of the following: pool-level support or path recall; context-level chain preservation under the budget actually used; and reader correctness under both natural and oracle or near-oracle contexts.

11.2 Report Diagnostics on Conditional Subsets

Diagnostics should respect the nested-event structure of equation 32. Faithfulness numbers should be reported on the chain-present subset; exposure ablations should hold membership fixed; fusion ablations should hold both membership and ordering fixed. Numbers that mix multiple conditions hide the channel of improvement and reduce reproducibility.

11.3 Always Report Budget and Cost

Agentic and iterative methods can improve accuracy by spending more tokens, more retrieval calls, or more verifier calls. Without budget-normalized numbers, comparison is unfair. A minimal cost report includes: total tokens per question (prompt and generation), retrieval calls per question, verifier calls per question, and end-to-end latency. Where possible, accuracy-cost curves should replace single operating points.

11.4 Distinguish Gold-Used and Oracle Support

Gold supporting facts in benchmarks are not necessarily the facts a system used. Two systems with the same answer can use different chains; one chain may be valid and the other may exploit

Table 14. Suggested minimum reporting for multi-hop RAG papers.

Reported quantity	Stage probed	Why it matters
Pool support/path recall at Top- N	Observability ($\mathcal{P}$)	Bounds the system; without it, downstream gains are unattributable.
Selected support recall at K	Selection ($\mathcal{S}$)	Reveals whether the budget preserves the chain.
Position of support in $\pi(E_K)$	Exposure ($\mathcal{O}$)	Detects lost-in-the-middle effects.
Oracle-evidence EM/F1	Fusion ($\mathcal{U}$)	Isolates reader capability from retrieval.
Deletion or counterfactual sensitivity	Faithfulness ($\mathcal{T}$)	Distinguishes causal use from parametric leakage.
Token, call, and latency cost	Efficiency / budget	Required for fair comparison across architectures.

a shortcut. Reporting whether the system’s actual evidence contains the gold supports, together with how often it differs, helps distinguish faithful and shortcut behavior.

11.5 Minimum Reporting Table

Table 14 distills the guidelines above into a minimal checklist. For each of the five component events in Proposition 1, the table names one quantity that a multi-hop RAG paper should report if it wishes to make an identifiable claim about that event, together with a brief statement of why the quantity is necessary. The table is deliberately short: a paper that reports fewer than these five numbers can still contribute to the field, but its main claim cannot in general be attributed to a specific bottleneck without further diagnostics.

11.6 Threats to the Latent-Chain View

For completeness, we list four limitations of our framing. First, C^* is set-valued, not unique; some questions admit many valid chains, and partial supervision systematically undercounts them. Second, latent-reasoning methods (Section 3.8) push chain inference into model internals, where external diagnostics do not apply without additional probes. Third, agentic systems may exhibit emergent behavior that is not well captured by any fixed factorization. Fourth, our success-factor decomposition (equation 32) is exact only as a chain-rule expansion over a particular nested event sequence; under parametric leakage and shortcut events, the relationship to end-to-end accuracy is one of upper bound rather than equality. We have flagged this in the relevant theorem and remarks, and recommend that empirical work treat the decomposition as a diagnostic tool rather than as a strict prediction.

12 CONCLUSION

This survey recasts multi-hop RAG as inference over a latent support chain observed through a noisy retrieval pool, budgeted context, evidence ordering, and generated answer. Five conditional coordinates— observability, selection preservation, exposure, fusion, and causal faithfulness— complement architectural taxonomies with a diagnostic account of failure. Equation 32 and Proposition 1 shows why grounded success is jointly limited and why empirical claims must identify the component event that changed. Conditional evidence utility remains a mechanism-level score for selection rather than an extra stage probability.

Evaluation should therefore be conditional: faithfulness on chain-absent examples conflates retrieval and attribution, while fusion on chain-missing examples conflates selection and reasoning. Reporting should expose all five factors even when a method targets only one. Promising systems will improve several factors without concealing a loss in another, for example through process-supervised retrieval, bridge-aware selection, or counterfactual training.

Agentic search reformulates evidence acquisition as policy learning; long-context readers change evidence granularity; latent-reasoning methods move fusion into hidden states. None removes the unobserved support chain, and each introduces distinct attribution or auditability risks. Naming that persistent object explicitly provides a stable basis for comparing the next generation of multi-hop RAG systems.

DATA AND ARTIFACT AVAILABILITY

A versioned companion artifact for this survey is publicly available at <https://github.com/TsingyuL/multi-hop-rag-survey>. The manuscript audits revision ac6d6d4. The repository contains machine-readable method and benchmark seeds, an imported working library, discovery queries, pipeline mappings, reporting guidance, figure sources, and validation and site-generation scripts. Record-level status fields separate reviewed taxonomy judgments from seeded, imported, and needs-review material; Appendix B reports the counts for this revision.

A CANONICAL RUNNING EXAMPLES

Throughout the manuscript we anchor the latent-chain formalism on a publicly verifiable bridge-type item from HotpotQA [212]. Table 15 records the abridged question, ordered support pages, and benchmark answer used consistently in the discussion. The example serves as a bridge-entity, ordering, exposure, and faithfulness probe in Sections 3, 4, 5, 7, and 9.

Table 15. Canonical running example anchored on HotpotQA. “Support pages” lists the ordered evidence pages, and the answer is the benchmark-provided reference string.

Dataset (split)	Question (abridged)	Ordered support pages	Reference answer
HotpotQA (dev)	Government position held by the woman who portrayed Corliss Archer in <i>Kiss and Tell</i> ?	(i) <i>Kiss and Tell (1945 film)</i> ; (ii) <i>Shirley Temple</i>	Chief of Protocol

The table is an explanatory anchor, not a newly released benchmark instance. Reproduction should use the original HotpotQA development data and its supporting-fact annotations.

B ARTIFACT COVERAGE AND CURATION STATE

Table 16 reports the counts that can be audited directly from the manuscript build and the public companion artifact. They describe different, overlapping layers and must not be read as a PRISMA-style funnel. Counts are pinned to companion revision ac6d6d4; later versions may expand the imported library or promote additional records after verification.

The imported library intentionally preserves broad adjacent work, including records that may later be excluded or reclassified. Promotion to the method catalog requires stable metadata, a defensible primary label, and a concrete pipeline mapping. Consequently, only reviewed records

Table 16. Auditable coverage layers in the manuscript and companion artifact. Method, benchmark, and intervention records are counted separately because they represent different units of analysis.

Layer	Count	Verification state	Role
Rendered bibliography	N_{bib} = 241	Compiled citation entries	Narrative breadth, including auxiliary background
Imported working library	N_{lib} = 229	Deduplicated import; per-record review pending	Broad discovery layer, not an endorsed taxonomy
Method catalog seed	N_{seed} = 23	Mixed seeded, reviewed, and needs-review states exposed per record	Released method-level evidence map
Benchmark seed	6	Curated benchmark records with stated caveats	Diagnostic benchmark map
Pipeline interventions	27	Stage–estimand mappings with status fields	Crosswalk from mechanisms to observable diagnostics

should support comparative taxonomy claims; neither the bibliography count nor the imported-library count is evidence of full taxonomy verification. The public coverage protocol documents the status definitions, discovery queries, adjudication rule, and correction policy.

C SCREENING PROTOCOL DETAILS

Search corpus and time window. The seed collection was assembled from canonical landing pages at ACL Anthology, OpenReview, arXiv, venue proceedings, and official project repositories, with updates required to consult more than one source where possible. The scope begins in January 2017 and extends through the artifact cut-off date, 11 July 2026. Because the review is living rather than prospectively registered, later releases must record a new cut-off date in `docs/coverage_protocol.md`.

Query templates. We combined the following clusters with boolean operators: *Task* $\in$ {multi-hop question answering, multi-hop reasoning, multi-hop retrieval, multi-step reasoning, compositional question answering, multi-hop fact verification, multi-step retrieval-augmented generation, agentic retrieval, multi-document reasoning}; *Mechanism* $\in$ {retrieval-augmented generation, retrieve-then-read, iterative retrieval, agentic search, graph-based reasoning, knowledge graph question answering, decomposition, chain of thought, process supervision, verification, citation generation, counterfactual evaluation, long-context reading}; *Family* $\in$ {dense retrieval, sparse retrieval, hybrid retrieval, reranker, planner, decomposer, reader, fusion-in-decoder, agent, tool use, reinforcement learning}.

Inclusion and exclusion. A paper was *in scope* if it (i) targets multi-hop tasks or multi-step reasoning over retrieved or graph evidence, (ii) introduces or analyzes a method, benchmark, or evaluation protocol relevant to at least one of our five estimands, and (iii) has a stable, publicly inspectable paper or proceedings record. A paper was out of scope if its multi-hop framing was incidental, if it studied pure single-hop QA without a mechanism that materially informs multi-hop retrieval, or if no stable artifact was available for inspection. Citation count was not an inclusion criterion, which avoids a publication-time bias against recent work.

Annotation. Each method promoted to the catalog seed records its primary architectural family, primary and secondary estimands, evidence source, pipeline stage, tasks, datasets, source URL, code

URL when verified, review status, and a short evidence note. Benchmark records separately expose diagnostics and caveats. These schemas support the released crosswalks used in Sections 5–7.

Reproducibility of the screening. The companion repository releases the discovery queries, catalog schemas, reviewed taxonomy, imported literature library, and validation scripts at <https://github.com/TsingyuL/multi-hop-rag-survey>. Reviewed records are kept separate from unverified candidates so that future updates can re-run the same discovery and validation workflow without conflating candidate coverage with the curated evidence base used in this survey.

D MATHEMATICAL NOTES

Granularity of evidence units

In Definition 1, evidence units e_i can be passages, sentences, table rows, image regions, code blocks, or knowledge-graph triples. The framework is applicable across evidence granularities, but the numerical estimands are granularity-dependent: refining a passage into sentences typically shortens the chain length required for correctness while altering the marginal retrieval probability, the budgeted selection probability, and the exposure distribution of each unit. Comparisons across granularities are therefore only meaningful when the evaluation unit is held fixed. Mixed-granularity chains (e.g., a sentence linked to a table row via a shared entity) are permitted; the bridge variable then ranges over a heterogeneous space that includes entities, dates, numeric matches, and image-text grounding links.

Bridge priors and observability

Chain observability $\mathcal{P}(q)$ marginalizes over chain orderings. Under the often-reasonable assumption that the underlying chain is a directed acyclic graph rather than a strict sequence, observability is properly redefined as

$$\mathcal{P}(q) = \Pr(\{e_1, \dots, e_L\} \subseteq \mathcal{P} \mid q),$$

and selection preservation, exposure, fusion, and faithfulness factor analogously. We keep the sequential form in the main text because all benchmarks we surveyed annotate ordered or unordered chains rather than richer DAG structures.

Decomposition under chain ambiguity

When multiple distinct chains support the same answer, the events in equation 32 should be read existentially: each stage requirement is defined as “there exists a valid minimal chain $C \in \mathfrak{C}^\star(q)$ for which the stage condition holds” (chain observable in L_N , chain preserved in E_K , chain exposed within depth r , chain correctly composed, chain causally used). This is the form used by oracle-style evaluators that count any complete supporting chain as correct. Under this existential reading, the chain-rule factorization of equation 32 remains an *exact* decomposition of the corresponding grounded-success event, and the bottleneck upper bound in Proposition 1 continues to apply because grounded success is still bounded above by the smallest of its five conditional stage factors.

Identifiability under partial supervision

Proposition 2 states that $\mathcal{U}$ is identifiable conditional on $(\mathcal{P}, \mathcal{S}, \mathcal{O}) = (1, 1, 1)$. Under partial supervision, where only a subset of bridge variables is annotated, identifiability weakens but does not collapse: $\mathcal{U}$ is identifiable up to permutation on annotated bridges and remains a lower bound on true fusion reliability on unannotated bridges, by an argument analogous to that of Yang et al. (2024) for hop-resolved attention attribution.

Table 17. Method-to-estimand mapping (selected, non-exhaustive).

Method family	$\mathcal{P}$	$\mathcal{S}$	$\mathcal{O}$	$\mathcal{U}$	$\mathcal{T}$
Sparse retrieval (BM25, SPLADE)	●	○			
Dense retrieval (DPR, Contriever, BGE)	●	○			
Hybrid retrieval (HyDE, query expansion)	●	○			
Iterative retrieval (MDR, Baleen, IRCot)	●	●			
Adaptive retrieval (FLARE, DRAGIN)	●	●	○		
Graph traversal (GRAFT-Net, PullNet, ToG)	●	●	○		
GraphRAG / HippoRAG / LightRAG	●	●	○		
Long-context readers (LongRAG, RAPTOR)			●	○	
Cross-encoder rerankers	○	●			
LLM-as-reranker (RankGPT, RankZephyr)	○	●			
Bridge-aware selectors (HopRetriever)		●	○		
Compression (LLMLingua, RECOMP, xRAG)		○	●		
Decomposition (DecompRC, Self-Ask, IRCot)	○	●		○	
Fusion-in-decoder, FiD-light				●	
CoT/ ToT / GoT reasoning				●	○
Latent reasoning (Quiet-STaR, Coconut)				●	
Self-verification, CoVe, CRITIC				○	●
PRM, RAG-Gym, MATH-SHEPHERD		○		●	●
Counterfactual evaluation, RARR					●
Citation generation (WebGPT, ALCE)					●
Agentic RAG (Search-R1, DeepRAG)	●	●	●	○	○

Log-scale decomposition and additive risk

Taking logs of equation 32 yields the exact identity

$$\log P(Y_g | q) = \log \theta_{\text{obs}} + \log \theta_{\text{sel}} + \log \theta_{\text{exp}} + \log \theta_{\text{fus}} + \log \theta_{\text{faith}}.$$

This form motivates an *additive shortfall* bookkeeping view: each conditional factor contributes a non-negative shortfall $-\log \theta_i$ to the total log deficit, and the largest shortfall identifies which factor is currently rate-limiting for q . The factors are conditional probabilities along a nested chain of events, not marginal probabilities of independent modules; the additive identity should therefore be read as a diagnostic ranking device rather than as a claim that the underlying stages are statistically independent. Practitioners can estimate each shortfall on the diagnostic subset that reaches the preceding event and rank remediation targets accordingly.

E METHOD CATALOGUE

Table 17 cross-references major method families discussed in Sections 5 and 6 against the five estimands they primarily move. The mapping is intentionally non-exclusive: many methods touch several estimands; we record the *strongest* effect with ● and *secondary* effects with ○. Empty cells indicate no first-order effect.

F WRITING AND REPORTING CHECKLIST

We summarize the reporting guidelines of Section 11 as a checklist that authors of new multi-hop RAG methods can apply before submission.

- **Estimand identification.** State which of the five estimands the proposed method primarily targets, and which it might inadvertently affect.
- **Conditional metrics.** Report retrieval coverage on chain-present examples, reader accuracy on chain-present examples, faithfulness on chain-present-and-attended examples, and corresponding chain-absent baselines.
- **Budget.** Report retrieved-token budget, prompt-token budget, total LLM calls, and wall-clock latency per query.
- **Used vs. gold support.** Report which gold passages were present in the context and which were attended to, distinguished from the full pool.
- **Process annotations.** If the benchmark provides path or process labels, report path-level metrics in addition to end-to-end metrics.
- **Counterfactual probes.** If feasible, report sensitivity to deletion or substitution of bridge evidence.
- **Failure analysis.** Provide a small table breaking down errors by latent-chain failure mode ($\mathcal{P}$ -loss, $\mathcal{S}$ -loss, $\mathcal{O}$ -loss, $\mathcal{U}$ -loss, $\mathcal{T}$ -loss).
- **Threats.** State the threats to validity of the latent-chain framing for the chosen benchmarks (Section 11).

G EXTENDED MAPPING: METHODS TO ESTIMANDS AND DATASETS

Table 18 provides a finer-grained mapping for selected recent methods, listing which datasets they were primarily evaluated on, which estimands they targeted, and which estimands they inadvertently affected ('spillover'). Spillovers are particularly important for fair comparison: a retrieval-quality improvement that also reduces exposure pressure should not be credited solely as a fusion improvement.

H CLAIM TEMPLATES

For each estimand we provide a template that authors can use to phrase empirical claims unambiguously. The templates are intentionally verbose; their purpose is to expose the conditioning assumptions that remain implicit in the common "*method X improves multi-hop accuracy by y points*" phrasing.

Observability claim. "On queries whose annotated chain is contained in pool $\mathcal{P}$ with probability at least α (under retrieval depth k), our method increases this probability from α to α' ; we do not claim improvement on queries whose chain is unobservable at depth k ."

Selection-utility claim. "Conditional on chain observability ($\mathcal{P} = 1$), our method increases the probability that all chain elements appear in the reader's context window from β to β' ; the increment is reported under fixed token budget B and reported reranker class."

Exposure claim. "Conditional on selection ($\mathcal{S} = 1$), our method increases the probability that all chain elements are attended to during reading from γ to γ' , measured by attention-weight masking experiments at threshold τ ."

Fusion claim. "Conditional on observability, selection, and exposure ($\mathcal{P} = \mathcal{S} = \mathcal{O} = 1$), our method increases the probability of producing the correct answer from δ to δ' . The remaining failures are attributed to fusion under the assumptions of Section 3."

Table 18. Extended method-to-estimand mapping with spillovers.

Method	Family	Primary	Spillover	Benchmarks
DPR	dense ret.	$\mathcal{P}$	$\mathcal{S}$	NQ, HotpotQA
Contriever	dense ret.	$\mathcal{P}$	$\mathcal{S}$	BEIR, HotpotQA
MDR	iterative	$\mathcal{P}, \mathcal{S}$	$\mathcal{O}$	HotpotQA, FEVER
Baleen	iterative	$\mathcal{P}, \mathcal{S}$	$\mathcal{O}$	HoVer, FEVEROUS
IRCoT	decomp.+ret.	$\mathcal{S}, \mathcal{U}$	$\mathcal{P}$	HotpotQA, 2WikiMQA
FLARE	adaptive	$\mathcal{S}$	$\mathcal{U}$	2WikiMQA, ASQA
GraphRAG	graph-RAG	$\mathcal{P}, \mathcal{S}$	$\mathcal{U}$	private, MuSiQue
HippoRAG	graph-RAG	$\mathcal{P}, \mathcal{S}$	$\mathcal{U}$	MuSiQue, 2WikiMQA
LongRAG	long-context	$\mathcal{O}$	$\mathcal{S}$	HotpotQA, MuSiQue
RAPTOR	long-context	$\mathcal{O}, \mathcal{U}$	$\mathcal{S}$	QuALITY, NarrativeQA
HopRetriever	bridge-aware	$\mathcal{S}$	$\mathcal{P}$	HotpotQA
LLMLingua	compression	$\mathcal{O}$	$\mathcal{S}$	MuSiQue, HotpotQA
RECOMP	compression	$\mathcal{O}$	$\mathcal{U}$	HotpotQA, NQ
RankGPT	reranker	$\mathcal{S}$	$\mathcal{P}$	BEIR, multi-hop
Self-RAG	adaptive	$\mathcal{S}, \mathcal{U}$	$\mathcal{T}$	ASQA, multi-hop
Self-Ask	decomp.	$\mathcal{S}, \mathcal{U}$	$\mathcal{P}$	2WikiMQA
RAG-Gym	PRM	$\mathcal{U}, \mathcal{T}$	$\mathcal{S}$	MuSiQue, HotpotQA
RARR	post-hoc	$\mathcal{T}$	$\mathcal{S}$	ASQA
CoVe	self-verify.	$\mathcal{U}, \mathcal{T}$	—	ASQA, multi-hop
ALCE	citation	$\mathcal{T}$	—	ASQA, ELI5
Search-R1	agentic-RL	$\mathcal{P}, \mathcal{S}, \mathcal{O}$	$\mathcal{U}$	HotpotQA, MuSiQue
R1-Searcher	agentic-RL	$\mathcal{P}, \mathcal{S}, \mathcal{U}$	$\mathcal{O}$	HotpotQA, MuSiQue
DeepRAG	agentic-RL	$\mathcal{P}, \mathcal{S}, \mathcal{U}$	$\mathcal{O}, \mathcal{T}$	MuSiQue, 2WikiMQA
PAR-RAG	planning	$\mathcal{S}, \mathcal{U}$	$\mathcal{P}$	HotpotQA, MuSiQue
VERITAS	verifier	$\mathcal{T}$	$\mathcal{U}$	ASQA, multi-hop
ReARTeR	re-arch.	$\mathcal{S}, \mathcal{U}$	$\mathcal{T}$	multi-hop

Faithfulness claim. “Conditional on a correct answer being produced, our method increases the probability that the answer is causally supported by the cited evidence chain from ϵ to ϵ' ; causal support is measured by counterfactual deletion sensitivity at perturbation budget ρ .”

Each template specifies (i) the conditioning event, (ii) the population of queries, (iii) the budget and protocol, and (iv) the auxiliary assumptions. We argue that adoption of this style would substantially reduce comparability problems across the multi-hop RAG literature.

REFERENCES

- [1] A. Asai, K. Hashimoto, H. Hajishirzi, R. Socher, and C. Xiong. 2020. Learning to retrieve reasoning paths over wikipedia graph for question answering. In *International Conference on Learning Representations*.
- [2] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*.
- [3] I. Augenstein, C. Lioma, D. Wang, et al. 2019. MultiFC: A real-world multi-domain dataset for evidence-based fact checking of claims. In *Empirical Methods in Natural Language Processing*.
- [4] Y. Bai, X. Lv, J. Zhang, et al. 2024. LongBench: A bilingual, multitask benchmark for long context understanding. In *Association for Computational Linguistics*.
- [5] M. Besta, N. Blach, A. Kubicek, et al. 2024. Graph of thoughts: Solving elaborate problems with large language models. In *AAAI Conference on Artificial Intelligence*.
- [6] E. Biran, D. Gottesman, S. Yang, M. Geva, and A. Globerson. 2024. Hopping too late: Exploring the limitations of large language models on multi-hop queries. In *Empirical Methods in Natural Language Processing*.
- [7] Y. Chang, M. Narang, H. Suzuki, et al. 2022. WebQA: Multihop and multimodal QA. In *Computer Vision and Pattern Recognition*.
- [8] D. Chen, A. Fisch, J. Weston, and A. Bordes. 2017. Reading wikipedia to answer open-domain questions. In *Association for Computational Linguistics*.
- [9] M. Chen et al. 2025. ReSearch: Reinforcement-learned search agents for open-domain multi-hop reasoning. *arXiv preprint* (2025).
- [10] W. Chen, M.-W. Chang, E. Schlinger, W. Wang, and W. W. Cohen. 2021. Open question answering over tables and text. In *International Conference on Learning Representations*.
- [11] W. Chen, X. Ma, X. Wang, and W. W. Cohen. 2023. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. In *Transactions on Machine Learning Research*.
- [12] W. Chen, H. Wang, J. Chen, et al. 2020. TabFact: A large-scale dataset for table-based fact verification. In *International Conference on Learning Representations*.
- [13] W. Chen, H. Zha, Z. Chen, et al. 2020. HybridQA: A dataset of multi-hop question answering over tabular and textual data. In *Findings of EMNLP*.
- [14] X. Chen, X. Gao, Y. Lin, et al. 2024. HiQA: A hierarchical contextual augmentation RAG for multi-document question answering. *arXiv preprint arXiv:2402.01767* (2024).
- [15] X. Cheng et al. 2024. xRAG: Extreme context compression for retrieval-augmented generation. *arXiv preprint arXiv:2405.13792* (2024).
- [16] Y. Cheng et al. 2025. DualRAG: A dual-policy retrieval framework for multi-hop question answering. *arXiv preprint* (2025).
- [17] Z. Chu et al. 2024. A survey of chain-of-thought reasoning: Advances, frontiers and future. *arXiv preprint arXiv:2309.15402* (2024).
- [18] B. Dalvi, P. Jansen, O. Tafjord, et al. 2021. Explaining answers with entailment trees. In *Empirical Methods in Natural Language Processing*.
- [19] P. Dasigi, K. Lo, I. Beltagy, et al. 2021. A dataset of information-seeking questions and answers anchored in research papers. In *NAACL*.
- [20] S. Dhuliawala, M. Komeili, J. Xu, et al. 2024. Chain-of-verification reduces hallucination in large language models. In *Findings of ACL*.
- [21] E. Dinan, S. Roller, K. Shuster, et al. 2019. Wizard of Wikipedia: Knowledge-powered conversational agents. In *International Conference on Learning Representations*.
- [22] Y. Dong et al. 2023. Hierarchical KGQA for multi-hop reasoning. In *ACL*.
- [23] D. Dua, S. Gupta, S. Singh, and M. Gardner. 2022. Successive prompting for decomposing complex questions. In *Empirical Methods in Natural Language Processing*.
- [24] D. Edge, H. Trinh, N. Cheng, et al. 2024. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130* (2024).
- [25] J. Eisenschlos, B. Dhingra, J. Bulian, B. Borschinger, and J. Boyd-Graber. 2021. Fool me twice: Entailment from Wikipedia gamification. In *NAACL*.
- [26] W. Fan et al. 2024. A survey on RAG meeting LLMs: Towards retrieval-augmented large language models. In *KDD*.
- [27] Y. Fang, S. Sun, Z. Gan, R. Pillai, S. Wang, and J. Liu. 2020. Hierarchical graph network for multi-hop question answering. In *EMNLP*.
- [28] Y. Feldman and R. El-Yaniv. 2019. Multi-hop paragraph retrieval for open-domain question answering. In *ACL*.
- [29] S. Feng, H. Wan, C. Gunasekara, et al. 2020. DialDoc: Doc2dial revisited as a dialog-based document grounding task. In *ACL DialDoc Workshop*.

- [30] J. Ferguson, M. Gardner, H. Hajishirzi, T. Khot, and P. Dasigi. 2020. IIRC: A dataset of incomplete information reading comprehension questions. In *EMNLP*.
- [31] T. Formal, B. Piwowarski, and S. Clinchant. 2021. SPLADE: Sparse lexical and expansion model for first stage ranking. In *SIGIR*.
- [32] L. Gao, Z. Dai, P. Pasupat, et al. 2023. RARR: Researching and revising what language models say, using language models. In *ACL*.
- [33] L. Gao, X. Ma, J. Lin, and J. Callan. 2023. Precise zero-shot dense retrieval without relevance labels. In *ACL*.
- [34] L. Gao, A. Madaan, S. Zhou, et al. 2023. PAL: Program-aided language models. In *ICML*.
- [35] T. Gao, H. Yen, J. Yu, and D. Chen. 2023. Enabling large language models to generate text with citations. In *EMNLP*.
- [36] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2023. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997* (2023). <https://arxiv.org/abs/2312.10997>
- [37] M. Geva, J. Bastings, K. Filippova, and A. Globerson. 2023. Dissecting recall of factual associations in auto-regressive language models. In *EMNLP*.
- [38] Z. Gou, Z. Shao, Y. Gong, et al. 2024. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *ICLR*.
- [39] S. Goyal et al. 2024. Think before you speak: Training language models with pause tokens. In *ICLR*.
- [40] S. Goyal, E. Zelikman, A. Gandhi, et al. 2024. Quiet-STaR: Language models can teach themselves to think before speaking. *arXiv preprint arXiv:2403.09629* (2024).
- [41] A. Gu and T. Dao. 2024. Mamba: Linear-time sequence modeling with selective state spaces. In *COLM*.
- [42] Y. Gu et al. 2022. A survey on complex factoid question answering with a focus on multi-hop reasoning. In *ACM Computing Surveys*.
- [43] Xinyan Guan, Jiali Zeng, Fandong Meng, Chunlei Xin, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Jie Zhou. 2025. DeepRAG: Thinking to retrieve step by step for large language models. *arXiv preprint arXiv:2502.01142* (2025). <https://arxiv.org/abs/2502.01142>
- [44] N. Guha, J. Nyarko, D. Ho, et al. 2024. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. In *NeurIPS Datasets and Benchmarks Track*.
- [45] J. Guo, J. Li, D. Li, et al. 2023. From images to textual prompts: Zero-shot visual question answering with frozen large language models. In *CVPR*.
- [46] Z. Guo, L. Xia, Y. Yu, et al. 2024. LightRAG: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779* (2024).
- [47] B. Gutiérrez, Y. Shu, Y. Gu, et al. 2024. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *NeurIPS*.
- [48] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M. Chang. 2020. REALM: Retrieval-augmented language model pre-training. In *ICML*.
- [49] D. Hannan, A. Jain, and M. Bansal. 2020. ManyModalQA: Modality disambiguation and QA over diverse inputs. In *AAAI*.
- [50] S. Hao et al. 2024. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769* (2024).
- [51] S. Hao, Y. Gu, H. Ma, et al. 2023. Reasoning with language model is planning with world model. In *EMNLP*.
- [52] G. He, Y. Lan, J. Jiang, W. X. Zhao, and J.-R. Wen. 2021. Improving multi-hop knowledge base question answering by learning intermediate supervision signals. In *WSDM*.
- [53] Q. He et al. 2024. Position-robust evidence reading for multi-hop RAG. *arXiv preprint* (2024).
- [54] W. He et al. 2024. WikiMixQA: A multimodal multi-hop QA benchmark. *arXiv preprint* (2024).
- [55] X. Ho, A. Nguyen, S. Sugawara, and A. Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *COLING*.
- [56] S. Hofstätter, J. Chen, K. Raman, and H. Zamani. 2023. FiD-light: Efficient and effective retrieval-augmented text generation. In *SIGIR*.
- [57] L. Huang, W. Yu, W. Ma, et al. 2023. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232* (2023).
- [58] Y. Huang and J. Huang. 2024. A survey on retrieval-augmented text generation for large language models. *arXiv preprint arXiv:2404.10981* (2024).
- [59] G. Izacard, M. Caron, L. Hosseini, et al. 2022. Unsupervised dense information retrieval with contrastive learning. In *Transactions on Machine Learning Research*.
- [60] G. Izacard and E. Grave. 2021. Leveraging passage retrieval with generative models for open-domain question answering. In *EACL*.

- [61] G. Izacard, P. Lewis, M. Lomeli, et al. 2023. Atlas: Few-shot learning with retrieval augmented language models. In *Journal of Machine Learning Research*.
- [62] S. Jeong, J. Baek, S. Cho, S. Hwang, and J. Park. 2024. Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In *NAACL*.
- [63] H. Jhamtani and P. Clark. 2020. Learning to explain: Datasets and models for identifying valid reasoning chains in multihop question-answering. In *EMNLP*.
- [64] S. Ji et al. 2026. A survey of retrieval, reasoning, and their integration for large language models. In *Transactions on Machine Learning Research*.
- [65] H. Jiang, Q. Wu, C.-Y. Lin, Y. Yang, and L. Qiu. 2023. LLMingua: Compressing prompts for accelerated inference of large language models. In *EMNLP*.
- [66] J. Jiang, K. Zhou, W. Zhao, and J.-R. Wen. 2023. UniKGQA: Unified retrieval and reasoning for solving multi-hop question answering over knowledge graph. In *ICLR*.
- [67] Y. Jiang, S. Bordia, Z. Zhong, et al. 2020. HoVer: A dataset for many-hop fact extraction and claim verification. In *Findings of EMNLP*.
- [68] Z. Jiang, X. Ma, and W. Chen. 2024. LongRAG: Enhancing retrieval-augmented generation with long-context LLMs. *arXiv preprint arXiv:2406.15319* (2024).
- [69] Z. Jiang, F. F. Xu, L. Gao, et al. 2023. Active retrieval augmented generation. In *EMNLP*.
- [70] Bowen Jin, Hansi Zeng, Zhenrui Yue, Dong Wang, Hamed Zamani, and Jiawei Han. 2025. Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516* (2025). <https://arxiv.org/abs/2503.09516>
- [71] D. Jin, E. Pan, N. Oufattole, et al. 2021. What disease does this patient have? A large-scale open domain question answering dataset from medical exams. In *Applied Sciences*.
- [72] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, and X. Lu. 2019. PubMedQA: A dataset for biomedical research question answering. In *EMNLP*.
- [73] V. Karpukhin, B. Oğuz, S. Min, et al. 2020. Dense passage retrieval for open-domain question answering. In *EMNLP*.
- [74] O. Khattab, C. Potts, and M. Zaharia. 2021. Baleen: Robust multi-hop reasoning at scale via condensed retrieval. In *NeurIPS*.
- [75] O. Khattab and M. Zaharia. 2020. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *SIGIR*.
- [76] T. Khot, H. Trivedi, M. Finlayson, et al. 2023. Decomposed prompting: A modular approach for solving complex tasks. In *ICLR*.
- [77] T. Kočiský, J. Schwarz, P. Blunsom, et al. 2018. The NarrativeQA reading comprehension challenge. In *Transactions of the ACL*.
- [78] S. Krishna et al. 2024. FRAMES: A benchmark for factuality, retrieval, and multi-hop RAG reasoning. *arXiv preprint* (2024).
- [79] J. Låla, O. O'Donoghue, A. Shtedritski, et al. 2023. PaperQA: Retrieval-augmented generative agent for scientific research. *arXiv preprint arXiv:2312.07559* (2023).
- [80] Y. Lan, G. He, J. Jiang, J. Jiang, W. Zhao, and J.-R. Wen. 2021. A survey on complex knowledge base question answering: Methods, challenges and solutions. In *IJCAI*.
- [81] K. Lee, M.-W. Chang, and K. Toutanova. 2019. Latent retrieval for weakly supervised open domain question answering. In *ACL*.
- [82] S. Lee et al. 2024. MMRA: Multimodal multi-hop retrieval-augmented benchmark. *arXiv preprint* (2024).
- [83] P. Lewis, E. Perez, A. Piktus, et al. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*.
- [84] D. Li et al. 2023. Towards verifiable text generation with evolving memory and self-reflection. In *EMNLP*.
- [85] D. Li, A. Sun, X. Tang, et al. 2023. A survey of large language model attribution. *arXiv preprint arXiv:2311.03731* (2023).
- [86] H. Li et al. 2024. A survey on decoupled retrieval-augmented generation. *arXiv preprint* (2024).
- [87] J. Li, X. Cheng, W. Zhao, J.-Y. Nie, and J.-R. Wen. 2023. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*.
- [88] S. Li et al. 2024. SubgraphRAG: Subgraph-grounded retrieval-augmented generation. *arXiv preprint* (2024).
- [89] S. Li, X. Li, L. Shang, et al. 2021. HopRetriever: Retrieve hops over wikipedia to answer complex questions. In *AAAI*.
- [90] X. Li et al. 2023. Reasoning-path-aware retrieval for multi-hop QA. In *ACL Findings*.
- [91] Z. Li et al. 2023. Selective decomposition for multi-hop reasoning. *arXiv preprint* (2023).
- [92] Z. Li et al. 2024. Self-checker for fact-checking factual claims by language models. In *ACL Findings*.
- [93] J. Liang, G. Su, H. Lin, Y. Wu, R. Zhao, and Z. Li. 2025. Reasoning RAG via system 1 or system 2: A survey on reasoning agentic retrieval-augmented generation for industry challenges. *arXiv preprint arXiv:2506.10408* (2025).

- [94] L. Liang et al. 2025. KAG: Boosting LLMs in professional domains via knowledge-augmented generation. *arXiv preprint arXiv:2409.13731* (2025).
- [95] H. Lightman, V. Kosaraju, Y. Burda, et al. 2024. Let’s verify step by step. In *ICLR*.
- [96] W. Lin et al. 2024. LLaVA-RAG: Multimodal retrieval augmented generation. *arXiv preprint* (2024).
- [97] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranajpe, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics* 12 (2024), 157–173. https://doi.org/10.1162/tac1_a_00638
- [98] N. F. Liu, T. Zhang, and P. Liang. 2023. Evaluating verifiability in generative search engines. In *EMNLP Findings*.
- [99] X. Liu, H. Yu, H. Zhang, et al. 2024. AgentBench: Evaluating LLMs as agents. In *ICLR*.
- [100] L. Luo, Y.-F. Li, G. Haffari, and S. Pan. 2024. Reasoning on graphs: Faithful and interpretable large language model reasoning. In *ICLR*.
- [101] Q. Lyu, S. Havaladar, A. Stein, et al. 2023. Faithful chain-of-thought reasoning. In *IJCNLP-AACL*.
- [102] X. Ma et al. 2024. Ex-FEVER: A multi-hop explainable fact verification dataset. *arXiv preprint arXiv:2310.09754* (2024).
- [103] X. Ma, L. Wang, N. Yang, F. Wei, and J. Lin. 2024. Fine-tuning LLaMA for multi-stage text retrieval. In *SIGIR*.
- [104] Y. Ma et al. 2024. Think-on-graph 2.0: Deep and interpretable LLM reasoning with knowledge graph-guided retrieval. *arXiv preprint* (2024).
- [105] A. Madaan, N. Tandon, P. Gupta, et al. 2023. Self-refine: Iterative refinement with self-feedback. In *NeurIPS*.
- [106] C. Malaviya, S. Lee, S. Chen, et al. 2024. ExpertQA: Expert-curated questions and attributed answers. In *NAACL*.
- [107] A. Mallen, A. Asai, V. Zhong, et al. 2023. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *ACL*.
- [108] P. Manakul, A. Liusie, and M. Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *EMNLP*.
- [109] V. Mavi, A. Jangra, and A. Jatowt. 2022. A survey on multi-hop question answering and generation. *arXiv preprint arXiv:2204.09140* (2022).
- [110] C. Mavromatis and G. Karypis. 2022. ReaRev: Adaptive reasoning for question answering over knowledge graphs. In *EMNLP Findings*.
- [111] C. Mavromatis and G. Karypis. 2024. GNN-RAG: Graph neural retrieval for large language model reasoning. *arXiv preprint arXiv:2405.20139* (2024).
- [112] J. Menick, M. Trebacz, V. Mikulik, et al. 2022. Teaching language models to support answers with verified quotes. *arXiv preprint arXiv:2203.11147* (2022).
- [113] S. Min, D. Chen, H. Hajishirzi, and L. Zettlemoyer. 2019. A discrete hard EM approach for weakly supervised question answering. In *EMNLP*.
- [114] S. Min, V. Zhong, L. Zettlemoyer, and H. Hajishirzi. 2019. Multi-hop reading comprehension through question decomposition and rescoring. In *ACL*.
- [115] M. Mohammadi et al. 2024. A survey of multi-hop machine reading comprehension. In *Knowledge-Based Systems*.
- [116] R. Nakano, J. Hilton, S. Balaji, et al. 2021. WebGPT: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332* (2021).
- [117] L. Nan, C. Hsieh, Z. Mao, et al. 2022. FeTaQA: Free-form table question answering. In *Transactions of the ACL*.
- [118] T. Nguyen et al. 2025. MARAG: Modality-aware retrieval-augmented generation for multimodal multi-hop QA. *arXiv preprint* (2025).
- [119] J. Ni, C. Qu, J. Lu, et al. 2022. Large dual encoders are generalizable retrievers. In *EMNLP*.
- [120] X. Ning, Z. Lin, Z. Zhou, et al. 2024. Skeleton-of-thought: Prompting LLMs for efficient parallel generation. In *ICLR*.
- [121] C. Niu, Y. Wu, J. Zhu, et al. 2024. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *ACL*.
- [122] L. Niu et al. 2025. A survey of knowledge graph-augmented reasoning for LLMs. *arXiv preprint* (2025).
- [123] R. Nogueira and K. Cho. 2019. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085* (2019). <https://arxiv.org/abs/1901.04085>
- [124] R. Nogueira, Z. Jiang, R. Pradeep, and J. Lin. 2020. Document ranking with a pretrained sequence-to-sequence model. In *Findings of EMNLP*.
- [125] NovelHopQA Authors. 2025. NovelHopQA: A long-context multi-hop question answering benchmark. *arXiv preprint* (2025).
- [126] L. Pan, X. Wu, X. Lu, et al. 2023. Fact-checking complex claims with program-guided reasoning. In *ACL*.
- [127] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu. 2024. Unifying large language models and knowledge graphs: A roadmap. In *IEEE Transactions on Knowledge and Data Engineering*.
- [128] Z. Pan et al. 2024. LLMingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. In *ACL Findings*.
- [129] J. S. Park, J. O’Brien, C. J. Cai, et al. 2023. Generative agents: Interactive simulacra of human behavior. In *UIST*.

- [130] S. G. Patil, T. Zhang, X. Wang, and J. E. Gonzalez. 2023. Gorilla: Large language model connected with massive APIs. *arXiv preprint arXiv:2305.15334* (2023).
- [131] A. Plaat et al. 2024. Reasoning with large language models, a survey. *arXiv preprint arXiv:2407.11511* (2024).
- [132] R. Pradeep, S. Sharifmoghaddam, and J. Lin. 2023. RankZephyr: Effective and robust zero-shot listwise reranking. *arXiv preprint arXiv:2312.02724* (2023).
- [133] O. Press, M. Zhang, S. Min, et al. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of EMNLP*.
- [134] S. Qiao, Y. Ou, N. Zhang, et al. 2023. Reasoning with language model prompting: A survey. In *ACL*.
- [135] Y. Qin, S. Hu, Y. Lin, et al. 2023. ToolBench: Benchmarking the tool-use capability of LLMs. *arXiv preprint* (2023).
- [136] Y. Qin, S. Liang, Y. Ye, et al. 2024. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *ICLR*.
- [137] L. Qiu, Y. Xiao, Y. Qu, et al. 2019. Dynamically fused graph network for multi-hop reasoning. In *ACL*.
- [138] S. Robertson and H. Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. In *Foundations and Trends in Information Retrieval*.
- [139] K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, and M. Zaharia. 2022. ColBERTv2: Effective and efficient retrieval via lightweight late interaction. In *NAACL*.
- [140] P. Sarthi, S. Abdullah, A. Tuli, et al. 2024. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *ICLR*.
- [141] A. Saxena, A. Tripathi, and P. Talukdar. 2020. Improving multi-hop question answering over knowledge graphs using knowledge base embeddings. In *ACL*.
- [142] T. Schick, J. Dwivedi-Yu, R. Dessi, et al. 2023. Toolformer: Language models can teach themselves to use tools. In *NeurIPS*.
- [143] M. Schlichtkrull, Z. Guo, and A. Vlachos. 2024. AVeriTeC: A dataset for real-world claim verification with evidence from the web. In *NeurIPS Datasets and Benchmarks Track*.
- [144] J. Schnitzler et al. 2024. MoreHopQA: More than multi-hop reasoning. *arXiv preprint arXiv:2406.13397* (2024).
- [145] B. Sel, A. Al-Tawaha, V. Khattar, R. Jia, and M. Jin. 2024. Algorithm of thoughts: Enhancing exploration of ideas in large language models. In *ICML*.
- [146] U. Shoham, M. Ivgi, A. Efrat, J. Berant, and O. Levy. 2023. ZeroSCROLLS: A zero-shot benchmark for long text understanding. In *EMNLP Findings*.
- [147] Z. Shao, Y. Gong, Y. Shen, et al. 2023. Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy. In *Findings of EMNLP*.
- [148] J. Shi, S. Cao, L. Hou, J. Li, and H. Zhang. 2021. TransferNet: An effective and transparent framework for multi-hop question answering over relation graph. In *EMNLP*.
- [149] Z. Shi et al. 2026. MultiCube-RAG: Multi-dimensional retrieval cubes for complex multi-hop RAG. *arXiv preprint* (2026).
- [150] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *NeurIPS*.
- [151] A. Singh, A. Ehtesham, S. Kumar, and T. T. Khoei. 2025. Agentic retrieval-augmented generation: A survey on agentic RAG. *arXiv preprint arXiv:2501.09136* (2025).
- [152] T. Song et al. 2025. R1-Searcher: Incentivizing the search capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2503.05592* (2025).
- [153] W. Su, Y. Tang, Q. Ai, Z. Wu, and Y. Liu. 2024. DRAGIN: Dynamic retrieval augmented generation based on the information needs of large language models. In *ACL*.
- [154] H. Sun, T. Bedrax-Weiss, and W. W. Cohen. 2019. PullNet: Open domain question answering with iterative retrieval on knowledge bases and text. In *EMNLP*.
- [155] H. Sun, B. Dhingra, M. Zaheer, K. Mazaitis, R. Salakhutdinov, and W. Cohen. 2018. Open domain question answering using early fusion of knowledge bases and text. In *EMNLP*.
- [156] J. Sun, C. Xu, L. Tang, et al. 2024. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *ICLR*.
- [157] W. Sun, L. Yan, X. Ma, et al. 2023. Is ChatGPT good at search? Investigating large language models as re-ranking agents. In *EMNLP*.
- [158] Zhongxiang Sun, Qipeng Wang, Weijie Yu, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Song Yang, and Han Li. 2025. ReARTeR: Retrieval-Augmented Reasoning with Trustworthy Process Rewarding. *arXiv preprint arXiv:2501.07861* (2025). <https://arxiv.org/abs/2501.07861>
- [159] A. Suri et al. 2024. DocHop: Multi-document multi-hop reasoning benchmark. *arXiv preprint* (2024).
- [160] O. Tafjord, B. Dalvi, and P. Clark. 2021. ProofWriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of ACL*.

- [161] A. Talmor and J. Berant. 2018. The web as a knowledge-base for answering complex questions. In *NAACL*.
- [162] A. Talmor, O. Yoran, A. Catav, et al. 2021. MultiModalQA: Complex question answering over text, tables and images. In *ICLR*.
- [163] R. Tang et al. 2024. Found-in-the-middle: Calibrating positional attention for long-context reading. In *ACL*.
- [164] Y. Tang and Y. Yang. 2024. MultiHop-RAG: Benchmarking retrieval-augmented generation for multi-hop queries. In *COLM*.
- [165] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal. 2018. FEVER: A large-scale dataset for fact extraction and verification. In *NAACL*.
- [166] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal. 2020. Is multihop QA in DiRe condition? Measuring and reducing disconnected reasoning. In *EMNLP*.
- [167] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal. 2022. MuSiQue: Multihop questions via single-hop question composition. In *Transactions of the ACL*.
- [168] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *ACL*.
- [169] G. Tsatsaronis, G. Balikas, P. Malakasiotis, et al. 2015. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. In *BMC Bioinformatics*.
- [170] M. Tu, K. Huang, G. Wang, J. Huang, X. He, and B. Zhou. 2020. Select, answer and explain: Interpretable multi-hop reading comprehension over multiple documents. In *AAAI*.
- [171] M. Tu, G. Wang, J. Huang, et al. 2019. Multi-hop reading comprehension across multiple documents by reasoning over heterogeneous graphs. In *ACL*.
- [172] S. Tworkowski, K. Staniszewski, M. Pacek, et al. 2023. Focused transformer: Contrastive training for context scaling. In *NeurIPS*.
- [173] N. Vu et al. 2025. Hierarchical and iterative planning for RAG: A multi-hop perspective. *arXiv preprint (2025)*.
- [174] D. Wadden, S. Lin, K. Lo, et al. 2020. Fact or fiction: Verifying scientific claims. In *EMNLP*.
- [175] H. Wan et al. 2024. SciQAG: A framework for generation of high quality multi-hop scientific QA pairs. *arXiv preprint (2024)*.
- [176] B. Wang, X. Yue, Y. Su, and H. Sun. 2024. Grokked transformers are implicit reasoners: A mechanistic journey to the edge of generalization. In *NeurIPS*.
- [177] G. Wang, Y. Xie, Y. Jiang, et al. 2024. Voyager: An open-ended embodied agent with large language models. In *Transactions on Machine Learning Research*.
- [178] H. Wang et al. 2025. ReAgent: A reasoning agent with retrieval and verification. *arXiv preprint (2025)*.
- [179] J. Wang et al. 2025. O1-Searcher: An open-source replication of OpenAI o1-style search reasoning. *arXiv preprint (2025)*.
- [180] L. Wang, W. Xu, Y. Lan, et al. 2023. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In *ACL*.
- [181] L. Wang, N. Yang, X. Huang, et al. 2022. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533 (2022)*.
- [182] M. Wang et al. 2024. Loong: Generating long contexts with linearizable transformers. *arXiv preprint (2024)*.
- [183] M. Wang et al. 2024. VisDoM: Visual document multi-hop QA. *arXiv preprint (2024)*.
- [184] P. Wang, L. Li, Z. Shao, et al. 2024. Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *ACL*.
- [185] X. Wang et al. 2024. Executable code actions elicit better LLM agents. In *ICML*.
- [186] X. Wang, J. Wei, D. Schuurmans, et al. 2023. Self-consistency improves chain of thought reasoning in language models. In *ICLR*.
- [187] Y. Wang, N. Lipka, R. A. Rossi, et al. 2024. Knowledge graph prompting for multi-document question answering. In *AAAI*.
- [188] J. Welbl, P. Stenetorp, and S. Riedel. 2018. Constructing datasets for multi-hop reading comprehension across documents. In *Transactions of the ACL*.
- [189] Y. Weng, M. Zhu, F. Xia, et al. 2023. Large language models are better reasoners with self-verification. In *Findings of EMNLP*.
- [190] T. Wolfson, M. Geva, A. Gupta, et al. 2020. Break it down: A question understanding benchmark. In *Transactions of the ACL*.
- [191] F. Wu et al. 2024. IterDRAGON: Iterative decomposition and retrieval for agentic multi-hop QA. *arXiv preprint (2024)*.
- [192] Jian Wu, Linyi Yang, Zhen Wang, Manabu Okumura, and Yue Zhang. 2024. CofCA: A step-wise counterfactual multi-hop QA benchmark. *arXiv preprint arXiv:2402.11924 (2024)*. <https://arxiv.org/abs/2402.11924>

- [193] Z. Xi et al. 2023. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864* (2023).
- [194] S. Xiao, Z. Liu, Y. Shao, and Z. Cao. 2022. RetroMAE: Pre-training retrieval-oriented language models via masked auto-encoder. In *EMNLP*.
- [195] S. Xiao, Z. Liu, P. Zhang, and N. Muennighoff. 2024. C-pack: Packaged resources to advance general Chinese embedding. In *SIGIR*.
- [196] J. Xie et al. 2024. Adaptive retrieval-augmented generation: A survey of adaptive strategies. *arXiv preprint* (2024).
- [197] G. Xiong et al. 2024. Benchmarking retrieval-augmented generation for medicine. In *Findings of ACL*.
- [198] Guangzhi Xiong, Qiao Jin, Xiao Wang, Yin Fang, Haolin Liu, Yifan Yang, Fangyuan Chen, Zhixing Song, Dengyu Wang, Minjia Zhang, Zhiyong Lu, and Aidong Zhang. 2025. RAG-Gym: Optimizing Reasoning and Search Agents with Process Supervision. *arXiv preprint arXiv:2502.13957* (2025). <https://arxiv.org/abs/2502.13957>
- [199] L. Xiong, C. Xiong, Y. Li, et al. 2021. Approximate nearest neighbor negative contrastive learning for dense text retrieval. In *ICLR*.
- [200] Wenhan Xiong, Xiang Lorraine Li, Srinivasan Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen tau Yih, Sebastian Riedel, Douwe Kiela, and Barlas Oğuz. 2021. Answering complex open-domain questions with multi-hop dense retrieval. In *International Conference on Learning Representations*. <https://arxiv.org/abs/2009.12756>
- [201] F. Xu, W. Shi, and E. Choi. 2024. RECOMP: Improving retrieval-augmented LMs with context compression and selective augmentation. In *ICLR*.
- [202] L. Xu et al. 2024. Document-level natural language inference for evidence verification. *arXiv preprint* (2024).
- [203] Y. Xu et al. 2024. Bridge-aware retrieval for multi-hop question answering. *arXiv preprint* (2024).
- [204] Z. Xu et al. 2023. ChatRAG-Bench: Benchmarking conversational retrieval-augmented generation. *arXiv preprint* (2023).
- [205] Zhichao Xu, Zongyu Wu, Yun Zhou, Aosong Feng, Kang Zhou, Sangmin Woo, Kiran Ramnath, Yijun Tian, Xuan Qi, Weikang Qiu, Lin Lee Cheong, and Haibo Ding. 2025. Beyond Correctness: Rewarding Faithful Reasoning in Retrieval-Augmented Generation. *arXiv preprint arXiv:2510.13272* (2025). <https://arxiv.org/abs/2510.13272>
- [206] J. Xue, M. Zheng, Y. Hu, et al. 2024. BadRAG: Identifying vulnerabilities in retrieval augmented generation of large language models. *arXiv preprint arXiv:2406.00083* (2024).
- [207] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling. 2024. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884* (2024).
- [208] S. Yang, E. Gribovskaya, N. Kassner, M. Geva, and S. Riedel. 2024. Do large language models latently perform multi-hop reasoning?. In *ACL*.
- [209] X. Yang et al. 2024. CRAG: Comprehensive RAG benchmark. In *NeurIPS Datasets and Benchmarks Track*.
- [210] Y. Yang et al. 2023. PRCA: Pluggable reward-driven contextual adapter for RAG. In *EMNLP*.
- [211] Z. Yang et al. 2022. MuGER: Multi-granularity evidence retrieval and reasoning for hybrid question answering. In *Findings of EMNLP*.
- [212] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. 2369–2380. <https://doi.org/10.18653/v1/D18-1259>
- [213] S. Yao, D. Yu, J. Zhao, et al. 2023. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS*.
- [214] S. Yao, J. Zhao, D. Yu, et al. 2023. ReAct: Synergizing reasoning and acting in language models. In *ICLR*.
- [215] M. Yasunaga, J. Kasai, R. Zhang, et al. 2019. ScisummNet: A large annotated corpus and content-impact models for scientific paper summarization with citation networks. In *AAAI*.
- [216] M. Yasunaga, H. Ren, A. Bosselut, P. Liang, and J. Leskovec. 2021. QA-GNN: Reasoning with language models and knowledge graphs for question answering. In *NAACL*.
- [217] H. Yen et al. 2024. Long-context evaluation for multi-hop reasoning. *arXiv preprint* (2024).
- [218] H. Yen, T. Gao, M. Hou, et al. 2024. HELMET: How to evaluate long-context language models effectively and thoroughly. *arXiv preprint arXiv:2410.02694* (2024).
- [219] D. Yu, C. Zhu, Y. Fang, et al. 2022. KG-FiD: Infusing knowledge graph in fusion-in-decoder for open-domain question answering. In *ACL*.
- [220] W. Yu, H. Zhang, X. Pan, et al. 2023. Chain-of-note: Enhancing robustness in retrieval-augmented language models. In *EMNLP*.
- [221] H. Zhang et al. 2025. PAR-RAG: Plan-and-revise retrieval-augmented generation. *arXiv preprint* (2025).
- [222] J. Zhang et al. 2022. Subgraph retrieval enhanced model for multi-hop knowledge base question answering. In *ACL*.
- [223] W. Zhang et al. 2024. ReST-MCTS*: LLM self-training via process reward guided tree search. In *NeurIPS*.
- [224] X. Zhang, A. Bosselut, M. Yasunaga, et al. 2022. GreaseLM: Graph reasoning enhanced language models for question answering. In *ICLR*.

- [225] X. Zhang, Y. Chen, S. Hu, et al. 2024. ∞ Bench: Extending long context evaluation beyond 100k tokens. In *ACL*.
- [226] Y. Zhang et al. 2024. Cumulative reasoning with large language models. *arXiv preprint* (2024).
- [227] Y. Zhang, Y. Li, L. Cui, et al. 2023. Siren’s song in the AI ocean: A survey on hallucination in large language models. *arXiv preprint arXiv:2309.01219* (2023).
- [228] Y. Zhang, R. Sun, Y. Chen, et al. 2024. Chain of agents: Large language models collaborating on long-context tasks. In *NeurIPS*.
- [229] Z. Zhang, A. Zhang, M. Li, and A. Smola. 2023. Automatic chain of thought prompting in large language models. In *ICLR*.
- [230] C. Zhao et al. 2021. Multi-step reasoning over unstructured text with beam dense retrieval. In *NAACL*.
- [231] P. Zhao et al. 2024. Retrieval-augmented generation for AI-generated content: A survey. *arXiv preprint arXiv:2402.19473* (2024).
- [232] R. Zhao et al. 2023. Verify-and-edit: A knowledge-enhanced chain-of-thought framework. In *ACL*.
- [233] Z. Zhong et al. 2023. Poisoning retrieval corpora by injecting adversarial passages. In *EMNLP*.
- [234] A. Zhou, K. Yan, M. Shlapentokh-Rothman, H. Wang, and Y.-X. Wang. 2024. Language agent tree search unifies reasoning, acting, and planning in language models. In *ICML*.
- [235] D. Zhou, N. Schärli, L. Hou, et al. 2023. Least-to-most prompting enables complex reasoning in large language models. In *ICLR*.
- [236] S. Zhou, F. F. Xu, H. Zhu, et al. 2024. WebArena: A realistic web environment for building autonomous agents. In *ICLR*.
- [237] A. Zhu, A. Cohen, A. Hu, K. Mahowald, and W. Yang. 2024. FanOutQA: A multi-hop, multi-document question answering benchmark for large language models. In *ACL*.
- [238] F. Zhu, W. Lei, Y. Huang, et al. 2021. TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance. In *ACL*.
- [239] H. Zhuang, Z. Qin, R. Jagerman, et al. 2023. RankT5: Fine-tuning T5 for text ranking with ranking losses. In *SIGIR*.
- [240] W. Zou, R. Geng, B. Wang, and J. Jia. 2025. PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *USENIX Security*.